

SMITHIAN FOLD THEORY V3

From One Sky to Cosmos

An Exact, Zero-Parameter and Machine-Closed Reconstruction of Astronomy and Cosmology
from Smithian Fold Theory

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Contents

Contents	2
An Exact, Zero-Parameter and Machine-Closed Reconstruction of Astronomy and Cosmology from Smithian Fold Theory	13
Abstract	13
Successor headline findings	14
Current status, evidence language and reader map	14
Corpus-wide terminology and evidence key	15
Three reading levels	15
Editorial change control	15
Authorship, access and why this work is public	16
The non-negotiable mathematical constitution	16
End-to-end empirical constitution	16
Dependency and ownership boundary	17
Primary source register	17
Family results	18
Source, observer, signal and geometry	18
Object identity, classification and catalogues	18
Radiation, motion and time	18
Stellar systems	19
Planetary systems and small bodies	19
Interstellar matter, enrichment and formation	19
Galaxies and populations	20
Compact, high-energy and multimessenger astronomy	20
Large-scale structure and expansion	20
Early universe, background and abundance	21
Dark sectors, horizons and unobserved claims	21
Inference, prediction and disciplinary handoffs	21
Complete claim-by-claim derivations	22
1. Astronomical source	22
Forward derivation	22
Falsification and controls	23
External evidence	23
What it means	23
Machine certificate	23
2. Astronomical observer	23
Forward derivation	23
Falsification and controls	24
External evidence	24
What it means	24
Machine certificate	25
3. Astronomical signal	25
Forward derivation	25

Falsification and controls	25
External evidence	26
What it means	26
Machine certificate	26
4. Angular position and reference frame	26
Forward derivation	26
Falsification and controls	27
External evidence	27
What it means	27
Machine certificate	27
5. Astronomical distance	27
Forward derivation	28
Falsification and controls	28
External evidence	28
What it means	29
Machine certificate	29
6. Survey selection and non-detection	29
Forward derivation	29
Falsification and controls	30
External evidence	30
What it means	30
Machine certificate	30
7. Astronomical object identity	30
Forward derivation	30
Falsification and controls	31
External evidence	31
What it means	32
Machine certificate	32
8. Astronomical classification	32
Forward derivation	32
Falsification and controls	33
External evidence	33
What it means	33
Machine certificate	33
9. Astronomical population	33
Forward derivation	33
Falsification and controls	34
External evidence	34
What it means	34
Machine certificate	35
10. Astronomical morphology	35
Forward derivation	35
Falsification and controls	35
External evidence	36
What it means	36

Machine certificate	36
11. Astronomical composition inference	36
Forward derivation	36
Falsification and controls	37
External evidence	37
What it means	37
Machine certificate	37
12. Astronomical catalogue	38
Forward derivation	38
Falsification and controls	38
External evidence	39
What it means	39
Machine certificate	39
13. Flux and luminosity	39
Forward derivation	39
Falsification and controls	40
External evidence	40
What it means	40
Machine certificate	40
14. Astronomical spectrum	40
Forward derivation	41
Falsification and controls	41
External evidence	41
What it means	42
Machine certificate	42
15. Astronomical redshift	42
Forward derivation	42
Falsification and controls	43
External evidence	43
What it means	43
Machine certificate	43
16. Astrometric and radial motion	43
Forward derivation	43
Falsification and controls	44
External evidence	44
What it means	45
Machine certificate	45
17. Astronomical variability	45
Forward derivation	45
Falsification and controls	46
External evidence	46
What it means	46
Machine certificate	46
18. Periodicity and transient distinction	46
Forward derivation	46

Falsification and controls	47
External evidence	47
What it means	47
Machine certificate	48
19. Stellar formation	48
Forward derivation	48
Falsification and controls	48
External evidence	49
What it means	49
Machine certificate	49
20. Stellar structure	49
Forward derivation	49
Falsification and controls	50
External evidence	50
What it means	50
Machine certificate	50
21. Stellar energy generation	51
Forward derivation	51
Falsification and controls	51
External evidence	52
What it means	52
Machine certificate	52
22. Stellar evolution	52
Forward derivation	52
Falsification and controls	53
External evidence	53
What it means	53
Machine certificate	53
23. Stellar population	53
Forward derivation	54
Falsification and controls	54
External evidence	54
What it means	55
Machine certificate	55
24. Stellar terminal state	55
Forward derivation	55
Falsification and controls	56
External evidence	56
What it means	56
Machine certificate	56
25. Planetary-system identity	56
Forward derivation	56
Falsification and controls	57
External evidence	57
What it means	58

Machine certificate	58
26. Astronomical orbit	58
Forward derivation	58
Falsification and controls	59
External evidence	59
What it means	59
Machine certificate	59
27. Orbital resonance and tidal state	59
Forward derivation	59
Falsification and controls	60
External evidence	60
What it means	60
Machine certificate	61
28. Planet and satellite classification	61
Forward derivation	61
Falsification and controls	61
External evidence	62
What it means	62
Machine certificate	62
29. Exoplanet observation	62
Forward derivation	62
Falsification and controls	63
External evidence	63
What it means	63
Machine certificate	63
30. Small-body population	64
Forward derivation	64
Falsification and controls	64
External evidence	65
What it means	65
Machine certificate	65
31. Interstellar-medium identity	65
Forward derivation	65
Falsification and controls	66
External evidence	66
What it means	66
Machine certificate	66
32. Astronomical dust	66
Forward derivation	67
Falsification and controls	67
External evidence	67
What it means	68
Machine certificate	68
33. Molecular cloud	68
Forward derivation	68

Falsification and controls	69
External evidence	69
What it means	69
Machine certificate	69
34. Cosmic chemical enrichment	69
Forward derivation	69
Falsification and controls	70
External evidence	70
What it means	71
Machine certificate	71
35. Environment-to-star-formation relation	71
Forward derivation	71
Falsification and controls	72
External evidence	72
What it means	72
Machine certificate	72
36. Stellar feedback	72
Forward derivation	72
Falsification and controls	73
External evidence	73
What it means	73
Machine certificate	74
37. Galaxy identity	74
Forward derivation	74
Falsification and controls	74
External evidence	75
What it means	75
Machine certificate	75
38. Galactic structure	75
Forward derivation	75
Falsification and controls	76
External evidence	76
What it means	76
Machine certificate	76
39. Galactic dynamics	76
Forward derivation	77
Falsification and controls	77
External evidence	77
What it means	78
Machine certificate	78
40. Luminosity-rotation scaling	78
Forward derivation	78
Falsification and controls	79
External evidence	79
What it means	79

Machine certificate	79
41. Galaxy evolution	79
Forward derivation	80
Falsification and controls	80
External evidence	80
What it means	81
Machine certificate	81
42. Groups, clusters and environment	81
Forward derivation	81
Falsification and controls	82
External evidence	82
What it means	82
Machine certificate	82
43. Compact-object identity	82
Forward derivation	82
Falsification and controls	83
External evidence	83
What it means	84
Machine certificate	84
44. Degenerate remnant classes	84
Forward derivation	84
Falsification and controls	85
External evidence	85
What it means	85
Machine certificate	85
45. Black-hole candidate and horizon evidence	85
Forward derivation	85
Falsification and controls	86
External evidence	86
What it means	86
Machine certificate	87
46. High-energy transient	87
Forward derivation	87
Falsification and controls	87
External evidence	88
What it means	88
Machine certificate	88
47. Multi-messenger association	88
Forward derivation	88
Falsification and controls	89
External evidence	89
What it means	89
Machine certificate	89
48. Compact-object population	90
Forward derivation	90

Falsification and controls	90
External evidence	91
What it means	91
Machine certificate	91
49. Large-scale statistical organisation	91
Forward derivation	91
Falsification and controls	92
External evidence	92
What it means	92
Machine certificate	92
50. Cosmic-web structure	92
Forward derivation	93
Falsification and controls	93
External evidence	93
What it means	94
Machine certificate	94
51. Cosmic expansion record	94
Forward derivation	94
Falsification and controls	95
External evidence	95
What it means	95
Machine certificate	95
52. Distance-ladder composition	95
Forward derivation	95
Falsification and controls	96
External evidence	96
What it means	96
Machine certificate	97
53. Gravitational-lensing observation	97
Forward derivation	97
Falsification and controls	97
External evidence	98
What it means	98
Machine certificate	98
54. Expansion-history inference	98
Forward derivation	98
Falsification and controls	99
External evidence	99
What it means	99
Machine certificate	99
55. Cosmic thermal history	99
Forward derivation	100
Falsification and controls	100
External evidence	101
What it means	101

Machine certificate	101
56. Cosmic microwave background	101
Forward derivation	101
Falsification and controls	102
External evidence	102
What it means	102
Machine certificate	102
57. Background acoustic structure	102
Forward derivation	103
Falsification and controls	103
External evidence	103
What it means	104
Machine certificate	104
58. Primordial abundance record	104
Forward derivation	104
Falsification and controls	105
External evidence	105
What it means	105
Machine certificate	105
59. Recombination and photon release	105
Forward derivation	105
Falsification and controls	106
External evidence	106
What it means	106
Machine certificate	107
60. Early perturbation record	107
Forward derivation	107
Falsification and controls	107
External evidence	108
What it means	108
Machine certificate	108
61. Dark-matter evidence class	108
Forward derivation	108
Falsification and controls	109
External evidence	109
What it means	109
Machine certificate	109
62. Dark-energy evidence class	109
Forward derivation	110
Falsification and controls	110
External evidence	111
What it means	111
Machine certificate	111
63. Cosmic horizon	111
Forward derivation	111

Falsification and controls	112
External evidence	112
What it means	112
Machine certificate	112
64. Cosmic topology claim	112
Forward derivation	113
Falsification and controls	113
External evidence	113
What it means	114
Machine certificate	114
65. Multiverse admissibility boundary	114
Forward derivation	114
Falsification and controls	115
External evidence	115
What it means	115
Machine certificate	115
66. Cosmic ultimate-fate inference	115
Forward derivation	115
Falsification and controls	116
External evidence	116
What it means	117
Machine certificate	117
67. Universal law and cosmic history distinction	117
Forward derivation	117
Falsification and controls	118
External evidence	118
What it means	118
Machine certificate	118
68. Astronomical model-observation boundary	118
Forward derivation	118
Falsification and controls	119
External evidence	119
What it means	119
Machine certificate	120
69. Standing astronomical prediction	120
Forward derivation	120
Falsification and controls	120
External evidence	121
What it means	121
Machine certificate	121
70. Unexplained astronomical observation	121
Forward derivation	121
Falsification and controls	122
External evidence	122
What it means	122

Machine certificate	122
71. Astrobiology and technosignature boundary	123
Forward derivation	123
Falsification and controls	123
External evidence	124
What it means	124
Machine certificate	124
72. Astronomy scientific handoffs	124
Forward derivation	124
Falsification and controls	125
External evidence	125
What it means	125
Machine certificate	125
Historical reconciliation	125
Limits and extension-open status	126
Full-field continuation roadmap	126
Reproduction	126
Rights, participation and scientific admission	126
Machine identities	126
Complete current-census successor supplement	127
Solar/radio unit magnitude-frequency release	127
Atomic Fold burst completion	127
Depth-seven planetary binary ladder	128
One-Fold lithium-seven depletion mechanism	129
Complete post-seal Earth/Astronomy prior comparison	129
Preceding-version abstract preserved for chronology	130
Registered source surface	131
Limitations and open frontier	131
Data and code availability	131
References and source registry	131
Successor machine-identity appendix	131
Lean 4 independent verification update	132
Verified result	132
Native theorem proof and artifact verification are different layers	132
Fail-closed custody event	132
What the result supports	133
Successor conclusion	133

An Exact, Zero-Parameter and Machine-Closed Reconstruction of Astronomy and Cosmology from Smithian Fold Theory

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in the same concept DOI lineage. The correction changes no scientific claim,

evidence classification, mathematical value, source identity or machine record.

Lean verification integration. The current SFT ordered census passed the

independent Lean 4 verification layer on 2 August 2026: 2,777 accepted claims,

898,902 candidate records and decisions, 11,108 controls, seventeen branches

and no reported issue. Lean natively proves the two-class operational root and

its unique survivor and constructs proof-bearing acceptance-gate certificates.

The remaining claim content is checked as the complete registered artifact

graph; that distinction is retained throughout this successor.

Abstract

This successor presents the complete current `astronomy_cosmology` paper surface: 77 live model-admitted claims, 19,712 generated candidates, 77 unique survivors and 308 passed controls. It integrates 5 claim section(s) not present in the preceding abstract's dated surface and remains complete only to this versioned registered census, with lawful extension open. The independent Lean 4 layer returned PASS for this surface as part of all 2,777 current claims.

Formal admission, implementation validation, observation, measurement, empirical support, confirmation, adverse evidence, unresolved evidence and publication status remain separate classifications. Lean does not reclassify an empirical result, repair an adverse observation or convert a later comparison into an earlier prediction. Lawful criticism, falsification and versioned extension remain open.

Successor headline findings

1. **Independent whole-model result.** Lean 4 returned `PASS` for 2,777/2,777 current claims, 898,902 candidates and decisions, 11,108 controls and all 17 branches, with no issue.
2. **Operational root.** The exact registered two-class grammar is formalised natively, and `presentedOccurrence` is proved to be its unique survivor without an imported or user-declared axiom in the exported theorem's axiom audit.
3. **Typed validation.** The root and generic acceptance implications are native Lean theorems; the remaining scientific claims are checked as complete registered artifacts. Empirical truth is not inferred from proof-assistant acceptance alone.
4. **Fail-closed custody.** A six-file byte-identity mismatch halted the first run. Exact registered bytes were restored, all 385 source bindings were re-audited, and the unchanged verifier then passed.
5. **Current paper surface.** This successor accounts for all 77 current claim(s) in its declared branch ownership boundary; 5 later claim section(s) are integrated in the successor supplement.
6. **Newly integrated results.** `SFT-ASTRO-SOLAR-RADIO-UNIT-RELEASE-002` - Solar/radio unit magnitude-frequency release; `SFT-ASTRO-ATOMIC-BURST-COMPLETION-002` - Atomic Fold burst completion; `SFT-ASTRO-PLANETARY-BINARY-LADDER-002` - Depth-seven planetary binary ladder; `SFT-ASTRO-LITHIUM-SEVEN-ONE-FOLD-DEPLETION-002` - One-Fold lithium-seven depletion mechanism; `SFT-ASTRO-VALIDATION-PRIOR-COMPLETE-FAMILY-002` - Complete post-seal Earth/Astronomy prior comparison.

These findings supplement rather than erase the branch-specific headline results retained below. Any adverse, corrected, unresolved, unavailable or chronology-bound result in the preceding scientific record remains governed by its current claim package.

Current status, evidence language and reader map

Status field	Current position
Paper and completion boundary	77 current claims in the declared branch surface. Completion is dated to the current registered census and remains open to lawful versioned extension.
Formal status	The Lean root theorem is kernel checked. Every live model claim has a current admitted receipt and passed the Lean artifact gates. Formal admission does not imply empirical confirmation.
Empirical status	Claim specific. Blind, non-blind, development-observed, holdout, favourable, adverse, null, missing, unavailable, disputed and unresolved distinctions remain exactly as recorded.
Chronology	Claim-specific registration, sealing, custody and observation order controls. Later evidence never becomes an earlier prediction without the registered chronology.
Publication status	Version 1.1.1 is published open access at 10.5281/zenodo.21761662 in the existing Zenodo concept DOI lineage.
Ownership	This paper retains its declared scientific boundary. Lean verification creates no new branch ownership and no application may select a law.
What is not claimed	Permanent closure of science, universal empirical confirmation, conversion of compatibility into confirmation, or superiority to every rival model.

Corpus-wide terminology and evidence key

Term	Reserved meaning in this paper
Theorem	A formally closed proposition at its declared grammar and dependency boundary.
Law	An admitted branch relation with its declared carrier, boundary, dependencies and extension rule.
Claim	The registered unit judged by the engine and bound to one immutable receipt.
Constitution	The rules governing admissible objects, derivations, evidence and publication; not an empirical result.
Derivation	Generation and elimination of the declared candidate space to its surviving structure.
Prediction	A consequence sealed before the matching target is released under the registered custody protocol.
Observation	A source-bound external record; it does not enter candidate generation.
Measurement	An instrument-, method-, condition- and uncertainty-bound observed value.
Reconstruction	A separately implemented regeneration or an explicitly identified inference of a retained state or history.
Exact numerical correspondence	Equality or registered interval relation between exact numerical objects at the declared boundary.
Structural correspondence	A post-derivation relation between forms without a claim of exact numerical prediction.
Boundary correspondence	Agreement restricted to the named interface, limit or ownership boundary.
Compatibility	Non-adverse but non-discriminating evidence.
Support	Relevant evidence that is not unique confirmation.
Confirmation	Used only when the current registered evidence protocol warrants that classification.
Validation	Successful execution of the named formal, computational or empirical test; its kind must be stated.
Adverse result	A registered test result that conflicts with the tested claim at its declared boundary.
Unresolved result	Required evidence remains unavailable, incomplete, disputed or insufficiently classified.
Implementation identity	The hash-bound identity of executable material; implementation success is not empirical confirmation.
Formal, empirical and publication status	Three independent classifications; none substitutes for another.
Foundational closure	Completion of the registered foundation only; not field-wide closure.
Field-wide closure	Completion of the named frozen or registered dated field census.
Current-evidence closure	Closure only to the evidence surface presently registered and preserved.
Extension openness	New questions may enter a later version without rewriting receipts or history.

Proof language is reserved for formal closure; derivation for generated structure; implementation for executable demonstration; prediction for a correctly sealed prospective consequence; observation and measurement for source-bound external records; reconstruction for separately regenerated or inferred states; correspondence for post-derivation relations; compatibility and support for non-unique evidence; confirmation only where the registered protocol authorises it; adverse for conflict; and unresolved where required evidence remains incomplete.

Three reading levels

1. **Conceptual paper.** The abstract, headline findings, branch narrative, major derivations, evidence synthesis, limitations and conclusion provide the readable scientific argument.
2. **Scientific audit layer.** Family and claim sections preserve exact status, dependencies, candidate and survivor counts, controls, chronology, sources, corrections, adverse evidence and reconciliation.
3. **Machine archive.** The repository packages preserve complete candidates, decisions, hashes, receipts, executable traces, source snapshots and certificates. They remain authoritative where prose abbreviates display.

Editorial change control

This successor improves currency, expression, typography and navigation. It does not change scientific meaning, chronology, evidence class, branch ownership, claim status or machine identity. Any conflict between authoritative records must be resolved by explicit supersession or reported to Maria Smith; prose alone cannot manufacture agreement.

Authorship, access and why this work is public

Maria Smith produced this work outside credentialed institutional access. That fact is not offered as a claim of personal exception. It is an indictment of how many capable minds science has excluded or silenced when money, credentials, paywalls, prestige and institutional permission are mistaken for rigour. The work must stand or fall on its visible derivations, executable tests, preserved adverse evidence and reproducible certificates-not on the biography of the author and not on the authority of a gatekeeper.

Ernos Labs is an open-source science movement and a standards designation. Anyone may inspect, criticize, reproduce, fork and extend the work under the stated licenses. Use of the **Ernos Labs** designation is conditional: the foundational theorem and declared constraints must remain visible; the protected engine and verification authority may not be weakened; candidate spaces and exclusions must be registered before outcomes; every alternative must be decided; failed, adverse, absent and unresolved evidence must be preserved; external evidence may not select a structural survivor; and criticism must remain distinguishable from admission. A proposal may be discussed openly without becoming part of the model. Admission requires the common machine-checkable route.

The paper is licensed **CC BY 4.0**. The platform code is licensed **Apache-2.0**. Maria Smith retains authorship, copyright and creative rights. Openness does not erase attribution, and attribution does not convert public science into private permission. Contact: Maria.Smith.Sftoe@gmail.com. Submissions and review: <https://discord.gg/ucwGryVxGr>. GitHub: <https://github.com/MettaMazza>.

The non-negotiable mathematical constitution

Every claim in this branch enters through one theorem-**There Is No Nothing**-and admitted Fold dependencies. There are no added axioms and no free or fitted parameters. Proof-state arithmetic is exact and finite. Conventional numerical zero is not an element of the proof carrier; where a user interface, catalogue or source writes 0, it is retained as a typed absence or missingness marker unless the source explicitly proves another meaning. Negative, irrational and imaginary quantities are not admitted as proof scalars. Completed infinities and ungenerated continua do not enter candidate generation. External decimals, signed coordinates and catalogue conventions are held labels in the measurement record; they compare the law but cannot become hidden premises.

For every claim, eight binary dimensions are generated literally: carrier, boundary, relation, record, evidence, provenance, generality and extension. This produces $2^8 = 256$ finite forms. Exactly one choice on each axis preserves the required distinction. The full Cartesian product is enumerated, every form is decided, and the single all-preserving form is independently reconstructed. Depth-independent closure means that adding one lawful finite source, epoch, channel, object or population member preserves prior identities and adverse rows while appending its own trace. It does not mean that the empirical universe is exhausted or that later observations are forbidden.

End-to-end empirical constitution

The complete 72-claim inventory, all 18,432 candidates, all predicted structural labels and all falsification conditions were sealed before external source identities or outcomes were opened. Primary custodians were then registered by purpose. Each target remained inaccessible to the capability-closed prediction process until its derivation seal existed. The evaluator opened the registered source record afterward, compared the result, ran deliberately changed controls, and preserved every transport failure and unfavourable row. The sealed engine rejected any missing dependency, incomplete census, non-unique survivor, failed control, changed source, failed independent reconstruction or empirical boundary violation.

The immutable engine seal is

sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a. The verification-authority seal is

sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8. They identify the unchanged common admission route; they are not substitutes for the derivations below.

Dependency and ownership boundary

The dependency chain is: foundational theorem → exact mathematics → information and computation → universal Physics → chemical identity and conservation → Earth observation handoff → Astronomy observation and history. Physics retains propagation, gravity, matter and constants. Chemistry retains species identity and transformation. Earth retains Earth-specific conditions. Engineering retains instrument implementation. Astronomy retains source identity, source-observer geometry, celestial populations, cosmic chronology and the custody of observational evidence. This division prevents double counting and prevents a telescope, simulation or catalogue from becoming a law generator.

Primary source register

- **ESA-GAIA-ARCHIVE-001 - European Space Agency.** Gaia ESA Archive. Registered purpose: observation_geometry, object_identity_classification, radiation_motion_time, stellar_systems, galaxies_populations. Locator: <https://gea.esac.esa.int/archive/>.
- **NASA-EXOPLANET-ARCHIVE-001 - NASA Exoplanet Science Institute.** NASA Exoplanet Archive. Registered purpose: planetary_small_body, stellar_systems. Locator: <https://exoplanetarchive.ipac.caltech.edu/>.
- **STSCI-MAST-001 - Space Telescope Science Institute.** Mikulski Archive for Space Telescopes. Registered purpose: observation_geometry, radiation_motion_time, stellar_systems, interstellar_enrichment_formation, galaxies_populations. Locator: <https://archive.stsci.edu/>.
- **CDS-SIMBAD-001 - Centre de Donnees astronomiques de Strasbourg.** SIMBAD astronomical database. Registered purpose: object_identity_classification, observation_geometry. Locator: <https://simbad.cds.unistra.fr/simbad/>.
- **MPC-DATA-SERVICES-001 - International Astronomical Union Minor Planet Center.** MPC Data Services. Registered purpose: planetary_small_body, observation_geometry. Locator: <https://data.minorplanetcenter.net/>.
- **SPARC-BTFR-001 - SPARC collaboration / Case Western Reserve University.** SPARC baryonic Tully-Fisher data. Registered purpose: galaxies_populations. Locator: https://astroweb.case.edu/SPARC/BTFR_Lelli2019.mrt.
- **GWOSC-PUBLIC-001 - LIGO Virgo KAGRA collaborations.** Gravitational Wave Open Science Center. Registered purpose: compact_high_energy_multimessenger. Locator: <https://gwosc.org/about/>.
- **ICECUBE-PUBLIC-DATA-001 - IceCube Collaboration.** IceCube public data releases. Registered purpose: compact_high_energy_multimessenger. Locator: <https://icecube.wisc.edu/science/data-releases/>.
- **NASA-HEASARC-001 - NASA High Energy Astrophysics Science Archive Research Center.** HEASARC data archive. Registered purpose: compact_high_energy_multimessenger, stellar_systems, galaxies_populations. Locator: <https://heasarc.gsfc.nasa.gov/docs/archive.html>.
- **NASA-LAMBDA-001 - NASA Goddard Space Flight Center.** Legacy Archive for Microwave Background Data Analysis. Registered purpose: large_scale_expansion, early_background_abundance, dark_horizon_unobserved. Locator: <https://lambda.gsfc.nasa.gov/>.
- **DESI-DR1-001 - Dark Energy Spectroscopic Instrument Collaboration.** DESI public Data Release 1. Registered purpose: galaxies_populations, large_scale_expansion, dark_horizon_unobserved. Locator: <https://data.desi.lbl.gov/doc/>.
- **SDSS-DATA-001 - Sloan Digital Sky Survey.** SDSS public data. Registered purpose: object_identity_classification, radiation_motion_time, galaxies_populations, large_scale_expansion. Locator: https://www.sdss.org/dr19/data_access/.
- **NASA-IRSA-001 - NASA Infrared Science Archive.** IRSA archives. Registered purpose: stellar_systems, planetary_small_body, interstellar_enrichment_formation, galaxies_populations. Locator: <https://irsa.ipac.caltech.edu/frontpage/>.
- **NASA-JPL-HORIZONS-001 - NASA Jet Propulsion Laboratory.** JPL Horizons system. Registered purpose: planetary_small_body. Locator: <https://ssd.jpl.nasa.gov/horizons/>.
- **NASA-ADS-001 - NASA Astrophysics Data System.** NASA ADS literature and data links. Registered purpose: inference_prediction_handoffs. Locator: <https://ui.adsabs.harvard.edu/>.

Family results

Source, observer, signal and geometry

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Astronomical source** (SFT-ASTRO-SOURCE-001): An astronomical source is a bounded carrier whose state produces or alters a registered signal along a declared path.
- **Astronomical observer** (SFT-ASTRO-OBSERVER-001): An observer is a source-located site, platform, instrument and clock relation, not an unlocated view.
- **Astronomical signal** (SFT-ASTRO-SIGNAL-001): A signal retains source, path, band, arrival, detector response and quality status.
- **Angular position and reference frame** (SFT-ASTRO-ANGULAR-POSITION-001): Angular position is a direction in a named frame and epoch with motion and uncertainty retained.
- **Astronomical distance** (SFT-ASTRO-DISTANCE-001): Distance is a method-bound source-observer relation whose ladder, calibration and range remain explicit.
- **Survey selection and non-detection** (SFT-ASTRO-SELECTION-001): Selection, detection, upper limit, non-detection, missing and outside-footprint status remain distinct.

Object identity, classification and catalogues

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Astronomical object identity** (SFT-ASTRO-OBJECT-001): Object identity is a reconstructible association of observations across epochs and channels.
- **Astronomical classification** (SFT-ASTRO-CLASSIFICATION-001): Classification retains measured features, method, alternatives and taxonomy version.
- **Astronomical population** (SFT-ASTRO-POPULATION-001): A population is a selection-bounded set with membership, volume, time, weights and censoring retained.
- **Astronomical morphology** (SFT-ASTRO-MORPHOLOGY-001): Morphology is a resolution- and band-bound organization of distinguishable components.
- **Astronomical composition inference** (SFT-ASTRO-COMPOSITION-001): Composition inference retains species evidence, excitation, method and alternatives.
- **Astronomical catalogue** (SFT-ASTRO-CATALOG-001): A catalogue is a versioned transformation with row provenance, flags, crossmatches and missingness.

Radiation, motion and time

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Flux and luminosity** (SFT-ASTRO-FLUX-LUMINOSITY-001): Received flux and source luminosity remain distinct through distance, geometry, band and attenuation records.
- **Astronomical spectrum** (SFT-ASTRO-SPECTRUM-001): A spectrum is an ordered calibrated signal with resolution, response, features and continuum retained.
- **Astronomical redshift** (SFT-ASTRO-REDSHIFT-001): Redshift is an emitted-to-observed feature relation; motion, gravity and expansion causes remain distinguishable.
- **Astrometric and radial motion** (SFT-ASTRO-MOTION-001): Motion retains frame, epochs, baseline, components and uncertainties.

- **Astronomical variability** (SFT-ASTRO-VARIABILITY-001): Variability is source change observed through a declared cadence and window with gaps retained.
- **Periodicity and transient distinction** (SFT-ASTRO-PERIOD-TRANSIENT-001): Periodic recurrence, quasi-periodicity and one-time transience remain distinct temporal classes.

Stellar systems

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Stellar formation** (SFT-ASTRO-STAR-FORMATION-001): Stellar formation is a source-dated assembly history from bound material to sustained stellar state.
- **Stellar structure** (SFT-ASTRO-STELLAR-STRUCTURE-001): A stellar state retains layers, mass, radius, composition, energy transport and observation boundary.
- **Stellar energy generation** (SFT-ASTRO-STELLAR-ENERGY-001): Stellar energy follows species through a complete reaction and loss ledger.
- **Stellar evolution** (SFT-ASTRO-STELLAR-EVOLUTION-001): Stellar evolution is a causal state sequence with initial conditions and mass loss retained.
- **Stellar population** (SFT-ASTRO-STELLAR-POPULATION-001): A stellar population retains selection, distance, extinction, binaries and completeness.
- **Stellar terminal state** (SFT-ASTRO-STELLAR-ENDPOINT-001): Terminal state links progenitor, transition, remnant and ejecta without erasing alternatives.

Planetary systems and small bodies

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Planetary-system identity** (SFT-ASTRO-PLANETARY-SYSTEM-001): A planetary system is a host-and-body organization with orbits, interactions and detection limits retained.
- **Astronomical orbit** (SFT-ASTRO-ORBIT-001): An orbit is a frame- and epoch-bound recurrent state sequence with perturbations and uncertainty.
- **Orbital resonance and tidal state** (SFT-ASTRO-RESONANCE-TIDE-001): Resonance is joint recurrence; tidal locking additionally retains dissipative history.
- **Planet and satellite classification** (SFT-ASTRO-PLANET-001): Planet, satellite and dwarf classifications retain dynamical context and measured features.
- **Exoplanet observation** (SFT-ASTRO-EXOPLANET-001): An exoplanet claim retains host, method, signal, selection, stellar activity and alternatives.
- **Small-body population** (SFT-ASTRO-SMALL-BODY-001): Small-body inference retains survey selection, orbit, activity, composition and completeness.

Interstellar matter, enrichment and formation

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Interstellar-medium identity** (SFT-ASTRO-ISM-001): The interstellar medium is a multiphase carrier with species, dust, fields and transfers retained.
- **Astronomical dust** (SFT-ASTRO-DUST-001): Dust inference retains grain properties, temperature, geometry and radiative method.
- **Molecular cloud** (SFT-ASTRO-MOLECULAR-CLOUD-001): A molecular cloud retains tracer, chemistry, density, motion and boundary.

- **Cosmic chemical enrichment** (SFT-ASTRO-CHEMICAL-ENRICHMENT-001): Enrichment is a source-to-ejecta-to-medium history retaining species, mixing and losses.
- **Environment-to-star-formation relation** (SFT-ASTRO-STAR-FORMATION-ENVIRONMENT-001): Star-formation relations retain region, tracer, time, selection and controls.
- **Stellar feedback** (SFT-ASTRO-FEEDBACK-001): Feedback follows source outputs into medium response with energy, momentum and species ledgers.

Galaxies and populations

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Galaxy identity** (SFT-ASTRO-GALAXY-001): A galaxy is a bound multi-component organization with environment and history retained.
- **Galactic structure** (SFT-ASTRO-GALACTIC-STRUCTURE-001): Galactic structure retains components, projection, kinematics and resolution.
- **Galactic dynamics** (SFT-ASTRO-GALACTIC-DYNAMICS-001): Galactic dynamics retain tracers, baryons, unseen inference, geometry and alternatives.
- **Luminosity-rotation scaling** (SFT-ASTRO-TULLY-FISHER-001): The pre-source rank-four relation is tested against complete source-bound galaxy rows and residuals.
- **Galaxy evolution** (SFT-ASTRO-GALAXY-EVOLUTION-001): Galaxy evolution distinguishes population sequence from individual causal history.
- **Groups, clusters and environment** (SFT-ASTRO-CLUSTER-LARGE-ENVIRONMENT-001): Cluster inference retains membership, gas, potential, selection and mass method.

Compact, high-energy and multimessenger astronomy

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Compact-object identity** (SFT-ASTRO-COMPACT-OBJECT-001): Compact-object identity retains source, environment, signals, method and alternatives.
- **Degenerate remnant classes** (SFT-ASTRO-WHITE-DWARF-NEUTRON-STAR-001): White-dwarf and neutron-star claims retain mass, radius, composition, rotation and field evidence.
- **Black-hole candidate and horizon evidence** (SFT-ASTRO-BLACK-HOLE-001): A black-hole claim retains dynamical mass, compactness, environment, messenger and alternatives.
- **High-energy transient** (SFT-ASTRO-TRANSIENT-001): A transient links source, event, localization, time, bands and afterglow with alternatives.
- **Multi-messenger association** (SFT-ASTRO-MULTIMESSENGER-001): Multi-messenger association retains localization, clocks, selection and background coincidence.
- **Compact-object population** (SFT-ASTRO-COMPACT-POPULATION-001): Compact populations retain detection selection, censoring and source history.

Large-scale structure and expansion

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Large-scale statistical organization** (SFT-ASTRO-COSMIC-HOMOGENEITY-001): Large-scale organization is a survey- and scale-bound tracer distribution.
- **Cosmic-web structure** (SFT-ASTRO-COSMIC-WEB-001): Web structure retains tracer, scale, method, selection and redshift.

- **Cosmic expansion record** (SFT-ASTRO-EXPANSION-RECORD-001): Expansion is a multi-probe distance-redshift record with calibration and covariance retained.
- **Distance-ladder composition** (SFT-ASTRO-DISTANCE-LADDER-001): A distance ladder is an explicit composition of anchors and rungs with covariance.
- **Gravitational-lensing observation** (SFT-ASTRO-LENSING-001): Lensing retains source, lens, observer, geometry, mass model and degeneracies.
- **Expansion-history inference** (SFT-ASTRO-EXPANSION-HISTORY-001): Expansion history is an ordered reconstruction from independently retained probes.

Early universe, background and abundance

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Cosmic thermal history** (SFT-ASTRO-THERMAL-HISTORY-001): Thermal history retains species, interaction, scale, temperature and evidence class by epoch.
- **Cosmic microwave background** (SFT-ASTRO-CMB-001): CMB evidence retains frequency, calibration, beam, masks, foregrounds and covariance.
- **Background acoustic structure** (SFT-ASTRO-ACOUSTIC-STRUCTURE-001): Acoustic structure retains harmonic features, covariance and extraction boundary.
- **Primordial abundance record** (SFT-ASTRO-PRIMORDIAL-ABUNDANCE-001): Primordial abundance inference retains species, sites, methods, depletion and stellar processing.
- **Recombination and photon release** (SFT-ASTRO-RECOMBINATION-001): Recombination links matter state, opacity and photon release with evidence boundaries.
- **Early perturbation record** (SFT-ASTRO-INITIAL-PERTURBATION-001): An early perturbation record retains scale, spectrum, covariance and transfer to later tracers.

Dark sectors, horizons and unobserved claims

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Dark-matter evidence class** (SFT-ASTRO-DARK-MATTER-EVIDENCE-001): Dark-matter evidence is a cross-probe residual with baryons and alternatives retained.
- **Dark-energy evidence class** (SFT-ASTRO-DARK-ENERGY-EVIDENCE-001): Dark-energy evidence retains distance, growth, calibration, covariance and alternatives.
- **Cosmic horizon** (SFT-ASTRO-COSMIC-HORIZON-001): A horizon is an observer- and epoch-bound causal access limit under admitted propagation.
- **Cosmic topology claim** (SFT-ASTRO-TOPOLOGY-001): Topology claims require observable repeated-pattern consequences and controls.
- **Multiverse admissibility boundary** (SFT-ASTRO-MULTIVERSE-001): A multiverse claim enters empirical science only through a distinct observable consequence sealed before data.
- **Cosmic ultimate-fate inference** (SFT-ASTRO-ULTIMATE-FATE-001): Ultimate fate is conditional continuation from current records and admitted laws, not an observed terminal state.

Inference, prediction and disciplinary handoffs

This family closes 6 nonduplicate foundational obligations. Each is current-evidence closed at its registered boundary and remains open to lawful extension.

- **Universal law and cosmic history distinction** (SFT-ASTRO-LAW-HISTORY-001): Universal law, contingent initial condition and source-dated history remain separate.

- **Astronomical model-observation boundary** (SFT-ASTRO-MODEL-OBSERVATION-001): A model output remains a conditional result compared with, not substituted for, observation.
- **Standing astronomical prediction** (SFT-ASTRO-STANDING-PREDICTION-001): A standing prediction is sealed before outcome and retains every adverse and missing row.
- **Unexplained astronomical observation** (SFT-ASTRO-UNEXPLAINED-001): Unexplained means a source-bound residual after declared controls, not evidence for a preferred unknown cause.
- **Astrobiology and technosignature boundary** (SFT-ASTRO-ASTROBIOLOGY-001): Life or technology claims require replicated signals and complete abiotic, instrumental and contamination controls.
- **Astronomy scientific handoffs** (SFT-ASTRO-ASTRO-HANDOFF-001): Astronomy consumes universal laws and exports source-dated constraints through explicit Physics, Engineering, Earth, Biology and Social boundaries.

Complete claim-by-claim derivations

The following sections are the paper-readable derivation record. The machine packages additionally preserve every one of the 256 candidates and every decision for each claim.

1. Astronomical source

Claim: SFT-ASTRO-SOURCE-001

Family: observation_geometry

Statement: An astronomical source is a bounded carrier whose state produces or alters a registered signal along a declared path.

Forward derivation

Carrier: emitting-or-interacting-source-carrier. **Relation:** source-state-to-signal-relation.

Retained record: identity-direction-distance-time-and-state-held. **Evidence boundary:** source-resolution-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
emitting-or-interacting-source-carrier__source-resolution-bounded__source-state-to-signal-relation__identity-direction-distance-time-and-state-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if source identity, direction, distance, time or state is erased.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- CDS-SIMBAD-001 - transport captured; snapshot
sha256:fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.

What it means

This law makes **astronomical source** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:565cccbfaff00ffa6dd9cd475e2289f9cc84f9ac4893d22ee17143533552797e.

Independent certificate:

sha256:3d8f3e27bb2c7d25363a541d5520a0145eacba97c04b34eac951f364f08cab2b. Engine receipt:
sha256:5473ed30f43a3ee1099bf327c08e196a9abcd888adb958b68d7c85d51681e958. Candidate count:
256. Survivor count: 1.

2. Astronomical observer

Claim: SFT-ASTRO-OBSERVER-001

Family: observation_geometry

Statement: An observer is a source-located site, platform, instrument and clock relation, not an unlocated view.

Forward derivation

Carrier: site-platform-instrument-observer-carrier. Relation: observer-to-record-relation.

Retained record: site-platform-clock-calibration-and-frame-held. Evidence boundary:
observer-frame-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,

SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
 SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-SOURCE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
site-platform-instrument-observer-carrier__observer-frame-bounded__observer-to-record
-relation__site-platform-clock-calibration-and-frame-held__evidence-class-explicit__r
oot-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if frame, clock, site or calibration changes without record.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

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sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- CDS-SIMBAD-001 - transport captured; snapshot
sha256:fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.

What it means

This law makes **astronomical observer** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:5cb3b2746ad8f0a5c180d9b1e92eadc20c7cf0dc09e86c5cd383007de876bd50.

Independent certificate:

sha256:c85c29c21faa35cbb4d71c03d513d5ffbedd0ec3cfe28874e374d303666c0930. Engine receipt: sha256:f714bc90e2a60620a46297a6839c7a5b3dbf02d7cd9e51f7b1c81071fb7d0d7a. Candidate count: 256. Survivor count: 1.

3. Astronomical signal

Claim: SFT-ASTRO-SIGNAL-001

Family: observation_geometry

Statement: A signal retains source, path, band, arrival, detector response and quality status.

Forward derivation

Carrier: source-path-detector-signal-carrier. Relation: propagation-and-detection-relation.

Retained record: emission-path-band-arrival-and-quality-held. Evidence boundary:

signal-channel-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-OBSERVER-001.

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Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
source-path-detector-signal-carrier__signal-channel-bounded__propagation-and-detection-relation__emission-path-band-arrival-and-quality-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if propagation, detector response or non-detection is omitted.

- false_premise - passed True: reject the generated form lacking the complete physical carrier

- `tampered_source` - passed True: reject a changed registered source identity
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External evidence

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- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- CDS-SIMBAD-001 - transport captured; snapshot
sha256:fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.

What it means

This law makes **astronomical signal** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:0f6f0e90e92f6674ca3dfa49c9e355681d6d2a9dc10fae571177321f9787afb9.

Independent certificate:

sha256:102bcc412ea7545492f737089158b2868a0a554bb5b463b13255d2a6a2bb9623. Engine receipt: sha256:c0417bce86220840e95fcec8e05c9888b85857274b388151c95c1a744e721b1e. Candidate count: 256. Survivor count: 1.

4. Angular position and reference frame

Claim: SFT-ASTRO-ANGULAR-POSITION-001

Family: `observation_geometry`

Statement: Angular position is a direction in a named frame and epoch with motion and uncertainty retained.

Forward derivation

Carrier: `direction-coordinate-frame-carrier`. Relation: `frame-transformation-relation`. Retained record: `epoch-origin-basis-motion-and-uncertainty-held`. Evidence boundary: `astrometric-frame-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,

SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-SIGNAL-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
direction-coordinate-frame-carrier__astrometric-frame-bounded__frame-transformation-r
elation__epoch-origin-basis-motion-and-uncertainty-held__evidence-class-explicit__roo
t-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject an unframed coordinate or an epoch-free moving position.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement /
primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- CDS-SIMBAD-001 - transport captured; snapshot
sha256:fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.

What it means

This law makes **angular position and reference frame** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:13defca3f4c6a212925722d48d5ce1f4e0de84a75c7fd8824fc566ea36f88edb.

Independent certificate:

sha256:ff66a96396bea270a578dbabe975d1f8a2257906b32adf59d807defb4838ad63. Engine receipt:
sha256:5baee62478fbcd17834102db63297cb4c72efcd2029f90652f9fc24d14588ae8. Candidate count:
256. Survivor count: 1.

5. Astronomical distance

Claim: SFT-ASTRO-DISTANCE-001

Family: observation_geometry

Statement: Distance is a method-bound source-observer relation whose ladder, calibration and range remain explicit.

Forward derivation

Carrier: source-observer-separation-carrier. **Relation:** method-specific-distance-inference.

Retained record: method-calibration-ladder-range-and-uncertainty-held. **Evidence boundary:** distance-method-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ANGULAR-POSITION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
source-observer-separation-carrier__distance-method-bounded__method-specific-distance
-inference__method-calibration-ladder-range-and-uncertainty-held__evidence-class-expl
icit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extr
a-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject if one distance method silently substitutes for another.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. **Captured sources:** 3; **unresolved transports retained:** 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- CDS-SIMBAD-001 - transport captured; snapshot
sha256:fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.

What it means

This law makes **astronomical distance** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:20ffa01ddc647a1445bcdaebd7fcedbf5835664239120675f6acef21496031cb.

Independent certificate:

sha256:13fabcb69b206253c29c490a33ebafa64e119c08664615d8b5d19f9e2a17b5e3. Engine receipt:

sha256:163fa182412a88c4e90f0006888a2d78ff4d2fb5d579fe938a5c6e521035aa54. Candidate count: 256. Survivor count: 1.

6. Survey selection and non-detection

Claim: SFT-ASTRO-SELECTION-001

Family: observation_geometry

Statement: Selection, detection, upper limit, non-detection, missing and outside-footprint status remain distinct.

Forward derivation

Carrier: target-opportunity-detection-status-carrier. Relation:

selection-and-detection-relation. Retained record:

footprint-cadence-threshold-masks-and-missingness-held. Evidence boundary:
survey-selection-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-DISTANCE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

target-opportunity-detection-status-carrier__survey-selection-bounded__selection-and-detection-relation__footprint-cadence-threshold-masks-and-missingness-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a non-detection or missing row becomes ordinary zero or confirmed absence.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- CDS-SIMBAD-001 - transport captured; snapshot
sha256:fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.

What it means

This law makes **survey selection and non-detection** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:0ee18d7570070e1263c2b53a4615453b48a38e51d6ddf05fa67de76b802eb06a.

Independent certificate:

sha256:241fc7a14a377add571071d1a922a384a90dcdd32058c86de97c2f2d2806adec. Engine receipt:
sha256:2c1a77d0fbc0a3ccd3180f637b9067c4f89fdd1fb2eaf2094bbfd74cf5bcb430. Candidate count:
256. Survivor count: 1.

7. Astronomical object identity

Claim: SFT-ASTRO-OBJECT-001

Family: object_identity_classification

Statement: Object identity is a reconstructible association of observations across epochs and channels.

Forward derivation

Carrier: cross-epoch-object-carrier. Relation: observation-association-relation. Retained record: positions-times-signals-links-and-alternatives-held. Evidence boundary: association-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-SELECTION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
cross-epoch-object-carrier__association-bounded__observation-association-relation__positions-times-signals-links-and-alternatives-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject if blended or ambiguous detections are forced into one object.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. **Captured sources:** 3; **unresolved transports retained:** 0. The post-seal consequence matched the registered structural label: True. **External evidence selected the survivor:** false.

- CDS-SIMBAD-001 - transport captured; snapshot
sha256:fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.
- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **astronomical object identity** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:7933a3cadb9ca1a48e829b5359b5ab4b3682357a96d6e1ad26d017e25e16ec04.

Independent certificate:

sha256:22128bc27aff2fc63ca7224a8596c43bb54f713757b05d3693825f39d09145d6. Engine receipt:

sha256:3aa8c2aa9a6e07c3a509e0f9ddfd46f777bbf74e489bf35ab967a0c7e165d08. Candidate count:

256. Survivor count: 1.

8. Astronomical classification

Claim: SFT-ASTRO-CLASSIFICATION-001

Family: object_identity_classification

Statement: Classification retains measured features, method, alternatives and taxonomy version.

Forward derivation

Carrier: feature-evidence-class-carrier. **Relation:** feature-to-class-relation. **Retained record:** features-method-probability-alternatives-and-version-held. **Evidence boundary:** classifier-and-taxonomy-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-SELECTION-001, SFT-ASTRO-OBJECT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

feature-evidence-class-carrier__classifier-and-taxonomy-bounded__feature-to-class-relation__features-method-probability-alternatives-and-version-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least

one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if a label replaces its evidence or an uncertain class becomes certain.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- CDS-SIMBAD-001 - transport captured; snapshot
sha256: `fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f`.
- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: `fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93`.
- SDSS-DATA-001 - transport captured; snapshot
sha256: `67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e`.

What it means

This law makes **astronomical classification** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256: `47440bf6c83f931457c48db4490812deaac6dce38b46a91f2df5a1b65d62c3cd`.

Independent certificate:

sha256: `4001adc5af38a0303fd9786200f61c7e81c9490896370de861253031a95cc7e7`. Engine receipt:
sha256: `38676430baf965fd0716b3e5386b3271651d850aff8c29d925e7fe0b9c831e56`. Candidate count:
256. Survivor count: 1.

9. Astronomical population

Claim: `SFT-ASTRO-POPULATION-001`

Family: `object_identity_classification`

Statement: A population is a selection-bounded set with membership, volume, time, weights and censoring retained.

Forward derivation

Carrier: `selection-bounded-object-set-carrier`. Relation: `membership-and-distribution-relation`. Retained record: `selection-volume-time-weights-and-censoring-held`. Evidence boundary: `population-sampling-bounded`.

Dependencies: `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`,
`SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`,
`SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`,
`SFT-MATH-PROBABILITY-STATISTICS-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`,
`SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-ENTROPY-UNCERTAINTY-001`,
`SFT-INFO-CONSERVATION-LOSS-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`,
`SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`, `SFT-PHYS-MEAS-UNCERTAINTY-001`,

SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-SELECTION-001, SFT-ASTRO-CLASSIFICATION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
selection-bounded-object-set-carrier__population-sampling-bounded__membership-and-distribution-relation__selection-volume-time-weights-and-censoring-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject population inference from favourable detections alone.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- CDS-SIMBAD-001 - transport captured; snapshot
sha256:fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.
- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **astronomical population** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:118634a6115dec00192524edf83a59a07f6bfdf633f4b36858656624bc0f3395.

Independent certificate:

sha256:bdd657bfd4ff44dc9bbc37bfebdal196b2d7d70fd21298cb2620d26fb6a6e3c0. Engine receipt: sha256:0a62514b2e54766ac4b30eedccd4c6ce170a42980849b27b1ddda55d3fa31441. Candidate count: 256. Survivor count: 1.

10. Astronomical morphology

Claim: SFT-ASTRO-MORPHOLOGY-001

Family: object_identity_classification

Statement: Morphology is a resolution- and band-bound organization of distinguishable components.

Forward derivation

Carrier: resolved-structure-carrier. **Relation:** component-adjacency-shape-relation. **Retained record:** resolution-band-orientation-distance-and-uncertainty-held. **Evidence boundary:** morphology-method-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-SELECTION-001, SFT-ASTRO-POPULATION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
resolved-structure-carrier__morphology-method-bounded__component-adjacency-shape-relation__resolution-band-orientation-distance-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject if resolution or projection is erased.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- CDS-SIMBAD-001 - transport captured; snapshot
sha256:fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.
- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **astronomical morphology** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:e88fe6a6c77d63f83bac0099b5092fb4a1438c2f09238322df9cd1049e2007d8.

Independent certificate:

sha256:4d77c06617da2c0ea2a9bd50df76468bc2f21ccf8919459a7309bae221ade1b. Engine receipt:
sha256:7e5ce0ee426488dc2e6d78aa5c6d1fca06d6cdccccfa3f0bd56b3eafa0fe44dc. Candidate count:
256. Survivor count: 1.

11. Astronomical composition inference

Claim: SFT-ASTRO-COMPOSITION-001

Family: `object_identity_classification`

Statement: Composition inference retains species evidence, excitation, method and alternatives.

Forward derivation

Carrier: `species-feature-source-carrier`. Relation: `spectral-or-particle-composition-relation`. Retained record: `species-lines-model-boundary-abundance-and-errors-held`. Evidence boundary: `composition-method-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,

SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-SELECTION-001, SFT-ASTRO-MORPHOLOGY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
species-feature-source-carrier__composition-method-bounded__spectral-or-particle-comp
osition-relation__species-lines-model-boundary-abundance-and-errors-held__evidence-cl
ass-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows_
_no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a line identification or abundance is detached from its method.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- CDS-SIMBAD-001 - transport captured; snapshot
sha256:fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.
- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **astronomical composition inference** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:97b806c8c263910ff4c3d0ad494c1381d9981221b5a79b9cd72c0f5f2e16b4c6.

Independent certificate:

sha256:4878bae7879c087c530b2f96c7487e1e7c9e1132faf6c55191b76d20c3d4a551. Engine receipt:
sha256:26cc87a10853fecab01b32b1fa0e693cfb8f6fd0cbec19091ee16fc2fdf5d17d. Candidate count:
256. Survivor count: 1.

12. Astronomical catalogue

Claim: SFT-ASTRO-CATALOG-001

Family: `object_identity_classification`

Statement: A catalogue is a versioned transformation with row provenance, flags, crossmatches and missingness.

Forward derivation

Carrier: `versioned-row-provenance-carrier`. **Relation:** `source-to-catalogue-transformation`.

Retained record: `query-version-columns-flags-crossmatches-and-missingness-held`. **Evidence**

boundary: `catalogue-release-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-SELECTION-001, SFT-ASTRO-COMPOSITION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
versioned-row-provenance-carrier__catalogue-release-bounded__source-to-catalogue-transformation__query-version-columns-flags-crossmatches-and-missingness-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. **Minimality:** `True`. **Named-form uniqueness:** `True`.

Falsification and controls

Falsification condition: Reject if catalogue version, query or quality filtering is hidden.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- CDS-SIMBAD-001 - transport captured; snapshot
sha256:fc1c9e0255db2941ebcfbe890cf3a5df1ca96ce285c8f8bc4d11a2087626d17f.
- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **astronomical catalogue** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:97174d95ba56d68090edccc04e6907baf0593c34bb3fe9fcd1b466964055235e.

Independent certificate:

sha256:0c92420b66a9497d20c6afdd1cc10dad772cfdb7aeec9d19fc8595a697035400. Engine receipt:
sha256:acdd861f1b9ddfb158859329b8062e970f19f77f497b673fe8caf0c4272f68bc. Candidate count:
256. Survivor count: 1.

13. Flux and luminosity

Claim: SFT-ASTRO-FLUX-LUMINOSITY-001

Family: `radiation_motion_time`

Statement: Received flux and source luminosity remain distinct through distance, geometry, band and attenuation records.

Forward derivation

Carrier: `source-path-received-radiation-carrier`. Relation: `flux-distance-luminosity-relation`. Retained record: `band-distance-geometry-extinction-and-time-held`. Evidence boundary: `radiometric-boundary-declared`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-CATALOG-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
source-path-received-radiation-carrier__radiometric-boundary-declared__flux-distance-
luminosity-relation__band-distance-geometry-extinction-and-time-held__evidence-class-
explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-
extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if apparent brightness is relabelled intrinsic luminosity.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **flux and luminosity** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:b19c3ed32c80f424415e8673ff8dda6605c3c09393a46c8f3c53898f8cd9a1dd.

Independent certificate:

sha256:1b7b2f2236e45abf9ccd59bf7988e7e485417753a0485bb211fba17e06e513f1. Engine receipt:
sha256:244626352d46e53db6f803dce4d8de4b5a9fc01a0c4a48c9ee7c74863b9f3f0a. Candidate count:
256. Survivor count: 1.

14. Astronomical spectrum

Claim: SFT-ASTRO-SPECTRUM-001

Family: radiation_motion_time

Statement: A spectrum is an ordered calibrated signal with resolution, response, features and continuum retained.

Forward derivation

Carrier: ordered-band-signal-carrier. **Relation:** wavelength-frequency-energy-feature-relation. **Retained record:** resolution-response-calibration-lines-and-continuum-held. **Evidence boundary:** spectral-method-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-CATALOG-001, SFT-ASTRO-FLUX-LUMINOSITY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
ordered-band-signal-carrier__spectral-method-bounded__wavelength-frequency-energy-
feature-relation__resolution-response-calibration-lines-and-continuum-held__evidence-
class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__
no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject an uncalibrated or selectively cropped feature set.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. **Captured sources:** 3; **unresolved transports retained:** 0. The post-seal consequence matched the registered structural label: True. **External evidence selected the survivor:** false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256: fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.

- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **astronomical spectrum** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:86ba6eeb0195e117e0f21b349976ca5a1666e8e5f1b2efcbd0150b49dcb56aa0.

Independent certificate:

sha256:a15145a02e581c0956ac90ef940158e7379a804df468c502edbd530448dc7011. Engine receipt:
sha256:3c2ae757898fd4ca1063639281403d361117113843e1a4bd69144ffa55655594. Candidate count:
256. Survivor count: 1.

15. Astronomical redshift

Claim: SFT-ASTRO-REDSHIFT-001

Family: radiation_motion_time

Statement: Redshift is an emitted-to-observed feature relation; motion, gravity and expansion causes remain distinguishable.

Forward derivation

Carrier: emitted-observed-feature-carrier. Relation: feature-displacement-relation. Retained record: feature-identity-frame-motion-gravity-expansion-and-error-held. Evidence boundary: redshift-cause-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-CATALOG-001, SFT-ASTRO-SPECTRUM-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

emitted-observed-feature-carrier__redshift-cause-bounded__feature-displacement-relation__feature-identity-frame-motion-gravity-expansion-and-error-held__evidence-class-ex

`plicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule`

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if measured redshift alone is treated as a unique distance or cause.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **astronomical redshift** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:ed2a05b04907340d5a36441d36f3e6ea31ff63ed5689d61e0e2012dd605ca85f.

Independent certificate:

sha256:35bfc901673ec988fb19ffa949a00854f8c17c3ad8c992741939f62ad35a845e. Engine receipt:
sha256:d3148d125f87843df676f13fcd10f8c9c05baa3afe48398d6743fa616d5606c2. Candidate count:
256. Survivor count: 1.

16. Astrometric and radial motion

Claim: `SFT-ASTRO-MOTION-001`

Family: `radiation_motion_time`

Statement: Motion retains frame, epochs, baseline, components and uncertainties.

Forward derivation

Carrier: `time-indexed-position-spectrum-carrier`. Relation:
`position-or-feature-change-relation`. Retained record:
`frame-epochs-baseline-components-and-errors-held`. Evidence boundary:
`motion-observation-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-CATALOG-001, SFT-ASTRO-REDSHIFT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
time-indexed-position-spectrum-carrier__motion-observation-bounded__position-or-feature-change-relation__frame-epochs-baseline-components-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject if apparent and physical motion are conflated.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. **Captured sources:** 3; **unresolved transports retained:** 0. The post-seal consequence matched the registered structural label: True. **External evidence selected the survivor:** false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **astrometric and radial motion** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:d3a960c27dc6800e5495d8cdee40bae9b4e9435e5a02a8fd0063472a90888a38.

Independent certificate:

sha256:0ba8e7ef3cbe87c6192cf84b713fdc99de580170846a34cda190a5816a8c8080. Engine receipt:

sha256:bbea1c97377efd636dc90e8df5d9963b2b6f55a59422efc302d0a0172897de8f. Candidate count:

256. Survivor count: 1.

17. Astronomical variability

Claim: SFT-ASTRO-VARIABILITY-001

Family: radiation_motion_time

Statement: Variability is source change observed through a declared cadence and window with gaps retained.

Forward derivation

Carrier: time-series-source-carrier. **Relation:** state-to-signal-change-relation. **Retained record:** cadence-window-band-errors-and-nondetections-held. **Evidence boundary:** time-series-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-CATALOG-001, SFT-ASTRO-MOTION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

time-series-source-carrier__time-series-bounded__state-to-signal-change-relation__cadence-window-band-errors-and-nondetections-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject if sampling aliases or missing intervals are erased.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **astronomical variability** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:f73f34c8d8bfaedff13e3ac0a5600eeddd28072872610c552df5fadfc94a40be.

Independent certificate:

sha256:310ff1e5bdfbf1862e8f7b9b9f56470f6b58f09944fd7b5fa7700f8c4eacd9c4. Engine receipt:
sha256:91d8d1c51df7727d0da50e2ebc76099744b8faca1c4dafd29fc99b962fc72aeb. Candidate count:
256. Survivor count: 1.

18. Periodicity and transient distinction

Claim: SFT-ASTRO-PERIOD-TRANSIENT-001

Family: `radiation_motion_time`

Statement: Periodic recurrence, quasi-periodicity and one-time transience remain distinct temporal classes.

Forward derivation

Carrier: `time-ordered-event-or-cycle-carrier`. **Relation:** `recurrence-or-one-event-relation`.

Retained record: `epochs-phase-window-baseline-and-alternatives-held`. **Evidence boundary:** `temporal-class-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,

SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-CATALOG-001, SFT-ASTRO-VARIABILITY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
time-ordered-event-or-cycle-carrier__temporal-class-bounded__recurrence-or-one-event-
relation__epochs-phase-window-baseline-and-alternatives-held__evidence-class-explicit
__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-ru
le
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a finite window manufactures a period or erases recurrence.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
 sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
 sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- SDSS-DATA-001 - transport captured; snapshot
 sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **periodicity and transient distinction** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:a15f56a691d4023dd5e50cf4fbedaadff6cc460c687b3ad771e8775e1d2af1bf.

Independent certificate:

sha256:c2b118aa01b580633cb3940fd3df7efea5fbd532e4c687cf6317519de49e28d3. Engine receipt: sha256:d4a7ebf3b590fe9908e73b46d032cbcb0e34ef99ad286ab3ff5644df05ac1b6a. Candidate count: 256. Survivor count: 1.

19. Stellar formation

Claim: SFT-ASTRO-STAR-FORMATION-001

Family: stellar_systems

Statement: Stellar formation is a source-dated assembly history from bound material to sustained stellar state.

Forward derivation

Carrier: bound-collapsing-gas-dust-carrier. Relation: assembly-heating-cooling-ignition-relation. Retained record: environment-mass-flow-time-and-evidence-held. Evidence boundary: formation-history-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-PERIOD-TRANSIENT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
bound-collapsing-gas-dust-carrier__formation-history-bounded__assembly-heating-cooling-ignition-relation__environment-mass-flow-time-and-evidence-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject if a final star is used as its own formation record.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **stellar formation** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:aff9065f6aff056a81855cc6cc00c0471ec33aaad49b5bc052237fd599cb571.

Independent certificate:

sha256:e762c8672780047e937e74efff63c138ea6ef388f194592a653480c9407d49b4. Engine receipt:
sha256:1687511f165df988b1a70cb12f64fda63bc0a8236daab845dd92d2f7a1723756. Candidate count:
256. Survivor count: 1.

20. Stellar structure

Claim: SFT-ASTRO-STELLAR-STRUCTURE-001

Family: stellar_systems

Statement: A stellar state retains layers, mass, radius, composition, energy transport and observation boundary.

Forward derivation

Carrier: layered-stellar-state-carrier. Relation: gravity-pressure-energy-transport-balance. Retained record: mass-radius-composition-temperature-luminosity-and-age-held. Evidence boundary: stellar-model-observation-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,

SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-PERIOD-TRANSIENT-001, SFT-ASTRO-STAR-FORMATION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
layered-stellar-state-carrier__stellar-model-observation-bounded__gravity-pressure-energy-transport-balance__mass-radius-composition-temperature-luminosity-and-age-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retain-s-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if model state is relabelled direct interior observation.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **stellar structure** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:a3997f5afb8a2b01da60edf201dc2f5140aeaa2718c3ad44c3522a8e3390119d.

Independent certificate:

sha256:54e7a20b55313ee7e0e5a15ab6b668abab7fd672ebcc20c89779e2b064f734ea. Engine receipt:
sha256:745723e85c03326d670183c817d64823618a9576ecd1efb2ace244d954efa0b5. Candidate count:
256. Survivor count: 1.

21. Stellar energy generation

Claim: SFT-ASTRO-STELLAR-ENERGY-001

Family: stellar_systems

Statement: Stellar energy follows species through a complete reaction and loss ledger.

Forward derivation

Carrier: fuel-reaction-product-radiation-carrier. **Relation:** reaction-network-to-output-ledger. **Retained record:** species-rates-losses-time-and-neutrino-photon-record-held. **Evidence boundary:** nuclear-network-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-PERIOD-TRANSIENT-001, SFT-ASTRO-STELLAR-STRUCTURE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
fuel-reaction-product-radiation-carrier__nuclear-network-bounded__reaction-network-to-
-output-ledger__species-rates-losses-time-and-neutrino-photon-record-held__evidence-c
lass-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows
__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject if missing energy or species is fitted away.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **stellar energy generation** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:af66d66442de0b714d0391fe86b6318428662926d24429e99f8fb3b8b545e863.

Independent certificate:

sha256:94ada775f758ba03641cf228ce143545c1981c0d944f1fb960e597d1bbd44656. Engine receipt:
sha256:677612669b5fd145ef82ad8bd915e8ef433e4a8c491ffc1b3467467d9f4db940. Candidate count:
256. Survivor count: 1.

22. Stellar evolution

Claim: SFT-ASTRO-STELLAR-EVOLUTION-001

Family: `stellar_systems`

Statement: Stellar evolution is a causal state sequence with initial conditions and mass loss retained.

Forward derivation

Carrier: `time-ordered-stellar-state-carrier`. Relation: `fuel-structure-transition-relation`. Retained record: `initial-mass-composition-environment-loss-and-time-held`. Evidence boundary: `evolution-track-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-PERIOD-TRANSIENT-001, SFT-ASTRO-STELLAR-ENERGY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
time-ordered-stellar-state-carrier__evolution-track-bounded__fuel-structure-transition-relation__initial-mass-composition-environment-loss-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a population ordering is silently treated as one star's history.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **stellar evolution** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:66c108b2959a933578e09e963e9f54983c00118989c9a3768ebbbf6cc6b9a3d1.

Independent certificate:

sha256:02257c3ab768120eb20d12b1c5efa8c2a6869f8e4f235dc42ae8faa9ec0c6b3b. Engine receipt:
sha256:452f1a61686abff87da13d1345308ad9722fd3c30e2db5ed71e63dd359a48f42. Candidate count:
256. Survivor count: 1.

23. Stellar population

Claim: SFT-ASTRO-STELLAR-POPULATION-001

Family: stellar_systems

Statement: A stellar population retains selection, distance, extinction, binaries and completeness.

Forward derivation

Carrier: selection-bounded-star-set-carrier. Relation: age-metallicity-mass-distribution-relation. Retained record: selection-distance-extinction-binaries-and-completeness-held. Evidence boundary: population-synthesis-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-PERIOD-TRANSIENT-001, SFT-ASTRO-STELLAR-EVOLUTION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
selection-bounded-star-set-carrier__population-synthesis-bounded__age-metallicity-mas
s-distribution-relation__selection-distance-extinction-binaries-and-completeness-held
__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retai
ns-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if unresolved binaries or selection effects are ignored.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.

- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **stellar population** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:048d32b7168b1f7fffd3213a961342a616910b7c1dc1d3a69081b2234eec2ad2.

Independent certificate:

sha256:e63ea43bb6eaa3b1a3117653309d670e5702818439949fb0d25311758ea2d48d. Engine receipt:
sha256:238a4d5f7c1962eea3c0039c054810fe25aa55a0f57218b3ac96885f99bce77a. Candidate count:
256. Survivor count: 1.

24. Stellar terminal state

Claim: SFT-ASTRO-STELLAR-ENDPOINT-001

Family: stellar_systems

Statement: Terminal state links progenitor, transition, remnant and ejecta without erasing alternatives.

Forward derivation

Carrier: progenitor-transition-remnant-carrier. Relation:
mass-composition-loss-collapse-relation. Retained record:
progenitor-event-remnant-ejecta-and-uncertainty-held. Evidence boundary:
endpoint-class-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-PERIOD-TRANSIENT-001, SFT-ASTRO-STELLAR-POPULATION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

progenitor-transition-remnant-carrier__endpoint-class-bounded__mass-composition-loss-collapse-relation__progenitor-event-remnant-ejecta-and-uncertainty-held__evidence-cla

ss-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if remnant class is inferred without progenitor and event boundaries.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- ESA-GAIA-ARCHIVE-001 - transport captured; snapshot
sha256:fd012f09514334064fd70b4191bc8c79d2f63c5fdeedc31a024c42c5f0cf0b93.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **stellar terminal state** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:9fe62de74937a175eebe6025cf41ad0ab9ba8ec4489138c425a2961fbefcfc98.

Independent certificate:

sha256:77800a391e1523af6300e79dec6f568d73005b769f6cb9852d1660fdd03babf9. Engine receipt:
sha256:cc3186110029ec8264d7cffb7583e93cdf8cb599853d2645374bc9c7377ad2f8. Candidate count:
256. Survivor count: 1.

25. Planetary-system identity

Claim: SFT-ASTRO-PLANETARY-SYSTEM-001

Family: planetary_small_body

Statement: A planetary system is a host-and-body organization with orbits, interactions and detection limits retained.

Forward derivation

Carrier: host-orbiting-body-joint-carrier. Relation: bound-orbit-and-interaction-relation.

Retained record: host-bodies-orbits-masses-epochs-and-errors-held. Evidence boundary:
system-detection-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,

SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
 SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
 SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
 SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
 SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-STELLAR-ENDPOINT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
host-orbiting-body-joint-carrier__system-detection-bounded__bound-orbit-and-interacti
on-relation__host-bodies-orbits-masses-epochs-and-errors-held__evidence-class-explici
t__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-r
ule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if candidates or unseen completeness are treated as confirmed bodies.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-EXOPLANET-ARCHIVE-001 - transport captured; snapshot
 sha256:78cd771cc5a4ed3f31f381048207300e09c5832c919181d49af67c0ac1e83d8c.
- MPC-DATA-SERVICES-001 - transport captured; snapshot
 sha256:107d2925b2ff051209d57c22181d1b485fd7f09b1ced9c5c79a49542668e2d77.
- NASA-JPL-HORIZONS-001 - transport captured; snapshot
 sha256:ae7b2f15acd48d137f7e261e8eef9a44faf6c72a655a28aeae5ee9a0a8bb928c.

What it means

This law makes **planetary-system identity** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:f36b76be1077d29fc769d1f1ab00606d6c124d775083225954c54b0ceccea027.

Independent certificate:

sha256:434327a0a8c1fb0f7eef01faa80e7fedbf3ba30a9807b0548fd323e0331ad6ca. Engine receipt:
sha256:c246e3a9b32c67d57ac4ec24cf388171bc7f56b403018ce48d1834def18b7d70. Candidate count:
256. Survivor count: 1.

26. Astronomical orbit

Claim: SFT-ASTRO-ORBIT-001

Family: planetary_small_body

Statement: An orbit is a frame- and epoch-bound recurrent state sequence with perturbations and uncertainty.

Forward derivation

Carrier: body-primary-state-sequence-carrier. **Relation:** bounded-recurrent-motion-relation.

Retained record: frame-epoch-state-elements-perturbations-and-errors-held. **Evidence boundary:** orbit-solution-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-STELLAR-ENDPOINT-001, SFT-ASTRO-PLANETARY-SYSTEM-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

body-primary-state-sequence-carrier__orbit-solution-bounded__bounded-recurrent-motion
-relation__frame-epoch-state-elements-perturbations-and-errors-held__evidence-class-e
xplicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-e
xtra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least

one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if fitted elements replace observations or an open trajectory is called bound.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- NASA-EXOPLANET-ARCHIVE-001 - transport captured; snapshot
sha256:78cd771cc5a4ed3f31f381048207300e09c5832c919181d49af67c0ac1e83d8c.
- MPC-DATA-SERVICES-001 - transport captured; snapshot
sha256:107d2925b2ff051209d57c22181d1b485fd7f09b1ced9c5c79a49542668e2d77.
- NASA-JPL-HORIZONS-001 - transport captured; snapshot
sha256:ae7b2f15acd48d137f7e261e8eef9a44faf6c72a655a28aeae5ee9a0a8bb928c.

What it means

This law makes **astronomical orbit** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:7b62450e2c54408cebec61a6742b4589ffe9255d89d29b92612b494064f5dad6.

Independent certificate:

sha256:7532424071c5951225ee7eeda4a039643700bd7619ec2815e03d65eaa3a4e2c. Engine receipt:
sha256:058bed515488bd9ed3e3f8e3a7f6cdbe218fef342910a86800cf6f40202f4be1. Candidate count:
256. Survivor count: 1.

27. Orbital resonance and tidal state

Claim: SFT-ASTRO-RESONANCE-TIDE-001

Family: `planetary_small_body`

Statement: Resonance is joint recurrence; tidal locking additionally retains dissipative history.

Forward derivation

Carrier: `coupled-orbit-spin-carrier`. Relation: `integer-recurrence-and-dissipation-relation`. Retained record: `periods-phases-bodies-history-and-errors-held`. Evidence boundary: `resonance-baseline-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,

SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-STELLAR-ENDPOINT-001, SFT-ASTRO-ORBIT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

coupled-orbit-spin-carrier__resonance-baseline-bounded__integer-recurrence-and-dissipation-relation__periods-phases-bodies-history-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject a near ratio without phase recurrence or a lock without history.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-EXOPLANET-ARCHIVE-001 - transport captured; snapshot
sha256:78cd771cc5a4ed3f31f381048207300e09c5832c919181d49af67c0ac1e83d8c.
- MPC-DATA-SERVICES-001 - transport captured; snapshot
sha256:107d2925b2ff051209d57c22181d1b485fd7f09b1ced9c5c79a49542668e2d77.
- NASA-JPL-HORIZONS-001 - transport captured; snapshot
sha256:ae7b2f15acd48d137f7e261e8eef9a44faf6c72a655a28aeae5ee9a0a8bb928c.

What it means

This law makes **orbital resonance and tidal state** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:a2c94e58be574845e88fa39a7bda07911bb4d45ec04953094315bfeed6f149ed.

Independent certificate:

sha256:83ad41de52b1a012f02ae2c6785ab09d5f4562e655af758c740080a9bbac8afe. Engine receipt:

sha256:7fb8371722eb7ced1d07e8e9617f2a368b04dd05adda017db98648d339afa87b. Candidate count:

256. Survivor count: 1.

28. Planet and satellite classification

Claim: SFT-ASTRO-PLANET-001

Family: planetary_small_body

Statement: Planet, satellite and dwarf classifications retain dynamical context and measured features.

Forward derivation

Carrier: body-orbit-mass-shape-context-carrier. Relation:

feature-and-dynamical-class-relation. Retained record:

host-orbit-mass-radius-composition-and-method-held. Evidence boundary:

classification-version-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-STELLAR-ENDPOINT-001, SFT-ASTRO-RESONANCE-TIDE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
body-orbit-mass-shape-context-carrier__classification-version-bounded__feature-and-dy
namical-class-relation__host-orbit-mass-radius-composition-and-method-held__evidence-
class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-row
s__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject if taxonomy supplies a physical law or uncertain mass is hidden.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-EXOPLANET-ARCHIVE-001 - transport captured; snapshot
sha256:78cd771cc5a4ed3f31f381048207300e09c5832c919181d49af67c0ac1e83d8c.
- MPC-DATA-SERVICES-001 - transport captured; snapshot
sha256:107d2925b2ff051209d57c22181d1b485fd7f09b1ced9c5c79a49542668e2d77.
- NASA-JPL-HORIZONS-001 - transport captured; snapshot
sha256:ae7b2f15acd48d137f7e261e8eef9a44faf6c72a655a28aeae5ee9a0a8bb928c.

What it means

This law makes **planet and satellite classification** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:6ca7cc9c45829a03f3f63a9b4f14c32d0d9af2734b2567346b248ba7f882853c.

Independent certificate:

sha256:179b1e4acc8322590db12527333eb316be24380622582a423afc39ef77b984e1. Engine receipt:
sha256:cc44ff010d51d2f18e183a9642d1c1f1bc3525ea658bd18e92973b259c69c90e. Candidate count:
256. Survivor count: 1.

29. Exoplanet observation

Claim: SFT-ASTRO-EXOPLANET-001

Family: `planetary_small_body`

Statement: An exoplanet claim retains host, method, signal, selection, stellar activity and alternatives.

Forward derivation

Carrier: `host-signal-planet-candidate-carrier`. Relation:
`transit-radial-imaging-or-timing-inference`. Retained record:
`method-selection-host-activity-period-and-errors-held`. Evidence boundary:
`detection-method-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,

SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-STELLAR-ENDPOINT-001, SFT-ASTRO-PLANET-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
host-signal-planet-candidate-carrier__detection-method-bounded__transit-radial-imagin
g-or-timing-inference__method-selection-host-activity-period-and-errors-held__evidenc
e-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-r
ows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if an instrumental or stellar signal is silently promoted to a planet.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement /
 primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-EXOPLANET-ARCHIVE-001 - transport captured; snapshot
 sha256:78cd771cc5a4ed3f31f381048207300e09c5832c919181d49af67c0ac1e83d8c.
- MPC-DATA-SERVICES-001 - transport captured; snapshot
 sha256:107d2925b2ff051209d57c22181d1b485fd7f09b1ced9c5c79a49542668e2d77.
- NASA-JPL-HORIZONS-001 - transport captured; snapshot
 sha256:ae7b2f15acd48d137f7e261e8eef9a44faf6c72a655a28aeae5ee9a0a8bb928c.

What it means

This law makes **exoplanet observation** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:ea5ead6a98089883080166720f37ddb856b12ac8fe7c3ff5266465668054f8e6.

Independent certificate:

sha256:843caea42accb3576284810f3fbdeb7d2240a1dbcabd8601ab3d3dff2af759a1. Engine receipt:
 sha256:b45e633c526545940df83343788a26f671c0061ddd899a9fae70251d37564314. Candidate count:
 256. Survivor count: 1.

30. Small-body population

Claim: SFT-ASTRO-SMALL-BODY-001

Family: planetary_small_body

Statement: Small-body inference retains survey selection, orbit, activity, composition and completeness.

Forward derivation

Carrier: selection-bounded-comet-asteroid-carrier. **Relation:** orbit-size-composition-population-relation. **Retained record:** survey-selection-orbits-activity-and-completeness-held. **Evidence boundary:** small-body-survey-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-STELLAR-ENDPOINT-001, SFT-ASTRO-EXOPLANET-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
selection-bounded-comet-asteroid-carrier__small-body-survey-bounded__orbit-size-composition-population-relation__survey-selection-orbits-activity-and-completeness-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject if detected objects are treated as the complete population.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-EXOPLANET-ARCHIVE-001 - transport captured; snapshot
sha256:78cd771cc5a4ed3f31f381048207300e09c5832c919181d49af67c0ac1e83d8c.
- MPC-DATA-SERVICES-001 - transport captured; snapshot
sha256:107d2925b2ff051209d57c22181d1b485fd7f09b1ced9c5c79a49542668e2d77.
- NASA-JPL-HORIZONS-001 - transport captured; snapshot
sha256:ae7b2f15acd48d137f7e261e8eef9a44faf6c72a655a28aeae5ee9a0a8bb928c.

What it means

This law makes **small-body population** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:19cae08f9d417aaccd353afa84f0880d9d08950e3d1d61efd9dc8c13b999c86f.

Independent certificate:

sha256:4630616a953f618776cbb45fce376333cd1132502480628ac2790624ff60ac6b. Engine receipt:
sha256:dc8a555a924c996226270750247dc628b5b4d7741b40acd27df07fd8015d85b9. Candidate count:
256. Survivor count: 1.

31. Interstellar-medium identity

Claim: SFT-ASTRO-ISM-001

Family: interstellar_enrichment_formation

Statement: The interstellar medium is a multiphase carrier with species, dust, fields and transfers retained.

Forward derivation

Carrier: multiphase-gas-dust-field-carrier. Relation: phase-and-transfer-network. Retained record: location-phase-species-field-scale-and-time-held. Evidence boundary: ism-observation-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-SMALL-BODY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
multiphase-gas-dust-field-carrier__ism-observation-bounded__phase-and-transfer-network__location-phase-species-field-scale-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if phases or unseen components are collapsed into one medium.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-IRSA-001 - transport captured; snapshot
sha256:1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **interstellar-medium identity** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:2e0084bbdf6cea58e5bc389d076e39f953de8804dba7bcc741f8882daa105b3d.

Independent certificate:

sha256:460f0407f41416cdce4b9f9832e47c32d6534f0350199301a6d1711d2b8ccad7. Engine receipt:
sha256:0cec1bfffce82ee255eab56babbd859767b0618e7791c555b77d4a3cc8a0467e4. Candidate count:
256. Survivor count: 1.

32. Astronomical dust

Claim: SFT-ASTRO-DUST-001

Family: interstellar_enrichment_formation

Statement: Dust inference retains grain properties, temperature, geometry and radiative method.

Forward derivation

Carrier: grain-composition-size-temperature-carrier. Relation: absorption-emission-scattering-relation. Retained record: composition-size-band-temperature-and-geometry-held. Evidence boundary: dust-model-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-SMALL-BODY-001, SFT-ASTRO-ISM-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
grain-composition-size-temperature-carrier__dust-model-bounded__absorption-emission-scattering-relation__composition-size-band-temperature-and-geometry-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if extinction and emission models are conflated.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- STSCI-MAST-001 - transport captured; snapshot
sha256: 6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.

- NASA-IRSA-001 - transport captured; snapshot
sha256:1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **astronomical dust** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:0c2e129fcb962f63506b440574274f57b43cede8b208645d3212445dbb7a6868.

Independent certificate:

sha256:3a98bd8a09d93ebecfbe23948092e24b3ad1a07b107a338132cf3f43b6c739fe. Engine receipt:
sha256:213ab20c5ab76aa3761d091100564d21726e2112b6030f48c4d30a319b19f67d. Candidate count:
256. Survivor count: 1.

33. Molecular cloud

Claim: SFT-ASTRO-MOLECULAR-CLOUD-001

Family: interstellar_enrichment_formation

Statement: A molecular cloud retains tracer, chemistry, density, motion and boundary.

Forward derivation

Carrier: molecule-gas-dust-bound-region-carrier. Relation:
chemistry-cooling-collapse-relation. Retained record:
species-density-temperature-motion-and-boundary-held. Evidence boundary:
tracer-method-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-SMALL-BODY-001, SFT-ASTRO-DUST-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

molecule-gas-dust-bound-region-carrier__tracer-method-bounded__chemistry-cooling-collapse-relation__species-density-temperature-motion-and-boundary-held__evidence-class-e

explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if one tracer is assumed to equal total material without a bridge.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-IRSA-001 - transport captured; snapshot
sha256:1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **molecular cloud** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:36286860fb3134180dfeae0f491055251b94c86c268c6ac01a3beecef5efd7bb7.

Independent certificate:

sha256:76689cd892a0727918bd008bd57690e38ba65658cb07489e733e72b5059ed00a. Engine receipt:
sha256:05313d27771f58964ebe43b714f0f74af495152b0ad8fc9362c3a0d5e9376fae. Candidate count:
256. Survivor count: 1.

34. Cosmic chemical enrichment

Claim: SFT-ASTRO-CHEMICAL-ENRICHMENT-001

Family: interstellar_enrichment_formation

Statement: Enrichment is a source-to-ejecta-to-medium history retaining species, mixing and losses.

Forward derivation

Carrier: source-ejecta-medium-generation-carrier. Relation:
nucleosynthesis-transfer-mixing-relation. Retained record:
species-yields-sources-time-mixing-and-losses-held. Evidence boundary:
enrichment-history-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-SMALL-BODY-001, SFT-ASTRO-MOLECULAR-CLOUD-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
source-ejecta-medium-generation-carrier__enrichment-history-bounded__nucleosynthesis-
transfer-mixing-relation__species-yields-sources-time-mixing-and-losses-held__evidenc
e-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-r
ows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject if abundance similarity proves unique provenance.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. **Captured sources:** 3; **unresolved transports retained:** 0. The post-seal consequence matched the registered structural label: True. **External evidence selected the survivor:** false.

- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-IRSA-001 - transport captured; snapshot
sha256:1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **cosmic chemical enrichment** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:c7db598e64b5fbc25cf346955a54a6f0e77bc79cd86c9aea3d4c0aa9cf6a55de.

Independent certificate:

sha256:96a7b74e90ec9eb47e5498a77dcbd07a931b2ba670704ebfecde71922425fc63. Engine receipt:

sha256:9cd403fe963694ee2817abb48d9e1c87cb7b38b49e61eb90c21eff8726bc85be. Candidate count:

256. Survivor count: 1.

35. Environment-to-star-formation relation

Claim: SFT-ASTRO-STAR-FORMATION-ENVIRONMENT-001

Family: interstellar_enrichment_formation

Statement: Star-formation relations retain region, tracer, time, selection and controls.

Forward derivation

Carrier: cloud-environment-young-star-carrier. Relation: condition-to-formation-rate-comparison. Retained record: mass-area-time-tracer-selection-and-controls-held. Evidence boundary: region-and-tracer-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-SMALL-BODY-001, SFT-ASTRO-CHEMICAL-ENRICHMENT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

cloud-environment-young-star-carrier__region-and-tracer-bounded__condition-to-formation-rate-comparison__mass-area-time-tracer-selection-and-controls-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least

one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject a universal rate from one tracer or region.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-IRSA-001 - transport captured; snapshot
sha256:1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **environment-to-star-formation relation** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:7c1212c0046c03473acf1b29db6ed405fa824c9392d36ddb63ff27f172a1086a.

Independent certificate:

sha256:96e562baf26a6e7ede30895e728187ccf22a0694adf5f66aa414052b302bf80b. Engine receipt:
sha256:28da28a0062a3873e96d08d96610ca8b8af37164f59251da7046dc4a27df6293. Candidate count:
256. Survivor count: 1.

36. Stellar feedback

Claim: SFT-ASTRO-FEEDBACK-001

Family: `interstellar_enrichment_formation`

Statement: Feedback follows source outputs into medium response with energy, momentum and species ledgers.

Forward derivation

Carrier: `star-output-medium-response-carrier`. Relation:
`radiation-wind-ejecta-transfer-relation`. Retained record:
`source-energy-momentum-species-geometry-and-time-held`. Evidence boundary:
`feedback-domain-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,

SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-SMALL-BODY-001, SFT-ASTRO-STAR-FORMATION-ENVIRONMENT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
star-output-medium-response-carrier__feedback-domain-bounded__radiation-wind-ejecta-t
ransfer-relation__source-energy-momentum-species-geometry-and-time-held__evidence-cla
ss-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__
no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if correlation alone establishes direction or conserved transfer.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.
- NASA-IRSA-001 - transport captured; snapshot
sha256:1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **stellar feedback** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:8df6e6a28b68fc8947dce890445c7cceb14985d5ddffc3e5c510f6c9aea09bb7.

Independent certificate:

sha256:d3a6a3293cd9fb267f7a6c25398812d1229b640d12be01f724fd727c827fad30. Engine receipt: sha256:1a44a2f07fadd9785f8d7e1b50f7eba9c43106c84cb5663b014133d5dbe5bedc. Candidate count: 256. Survivor count: 1.

37. Galaxy identity

Claim: SFT-ASTRO-GALAXY-001

Family: galaxies_populations

Statement: A galaxy is a bound multi-component organization with environment and history retained.

Forward derivation

Carrier: stars-gas-dust-dark-joint-carrier. Relation: bound-structure-and-history-relation.

Retained record: components-distance-frame-environment-and-time-held. Evidence boundary:

galaxy-observation-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-FEEDBACK-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
stars-gas-dust-dark-joint-carrier__galaxy-observation-bounded__bound-structure-and-history-relation__components-distance-frame-environment-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if morphology alone supplies identity or causal history.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.
- NASA-IRSA-001 - transport captured; snapshot
sha256:1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.

What it means

This law makes **galaxy identity** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:acfade8fe9bf241e079008cee407de9767f5b57ad772580d98807b0432010f4b.

Independent certificate:

sha256:ed5126efdde5a95a40e1f80c63af6cbd49eb68835b69d6ac1ac89523b82974a9. Engine receipt:

sha256:39520fdbde70a6f44e6e576a76668c0770b8b6e5a092add65a5b688f55d7135c. Candidate count: 256. Survivor count: 1.

38. Galactic structure

Claim: SFT-ASTRO-GALACTIC-STRUCTURE-001

Family: galaxies_populations

Statement: Galactic structure retains components, projection, kinematics and resolution.

Forward derivation

Carrier: resolved-component-network-carrier. **Relation:** spatial-kinematic-component-relation.

Retained record: band-resolution-orientation-distance-and-errors-held. **Evidence boundary:** structure-method-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,

SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-FEEDBACK-001, SFT-ASTRO-GALAXY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
resolved-component-network-carrier__structure-method-bounded__spatial-kinematic-component-relation__band-resolution-orientation-distance-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if projected light is treated as complete mass structure.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.
- NASA-IRSA-001 - transport captured; snapshot
sha256:1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.

What it means

This law makes **galactic structure** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:9605d9349b1444c9e48b28f96d49b85b9e3438654c18dbbac6c57f2b5ea21962.

Independent certificate:

sha256:dae64c6f8a5ac48a95efac6b37826303cfa8b2fad92cd619ce0a15012d104a38. Engine receipt:
sha256:8c55b5bfe11c3d144090c351c4852344af828c2ec7fffe05e207fc6d015617f5. Candidate count:
256. Survivor count: 1.

39. Galactic dynamics

Claim: SFT-ASTRO-GALACTIC-DYNAMICS-001

Family: galaxies_populations

Statement: Galactic dynamics retain tracers, baryons, unseen inference, geometry and alternatives.

Forward derivation

Carrier: tracer-potential-motion-carrier. **Relation:** mass-distribution-to-motion-relation.

Retained record: tracers-distance-inclination-baryons-dark-and-errors-held. **Evidence boundary:** dynamical-model-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-FEEDBACK-001, SFT-ASTRO-GALACTIC-STRUCTURE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
tracer-potential-motion-carrier__dynamical-model-bounded__mass-distribution-to-motion
-relation__tracers-distance-inclination-baryons-dark-and-errors-held__evidence-class-
explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-
extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject if a rotation curve uniquely identifies an untested mass model.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. **Captured sources:** 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.
- NASA-IRSA-001 - transport captured; snapshot
sha256:1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.

What it means

This law makes **galactic dynamics** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:beb7eaea295a68f5db3c1958329cf240d595329fa3e0a413b3111a71373379b6.

Independent certificate:

sha256:416dd1ee73a40efa2a237dcba4d54ca98f25e1e736357b74dc34286c90c01077. Engine receipt:
sha256:09221a0166047713578ad34a5ba48d2748ab3b857e0ac7b181766c37110d408d. Candidate count:
256. Survivor count: 1.

40. Luminosity-rotation scaling

Claim: SFT-ASTRO-TULLY-FISHER-001

Family: galaxies_populations

Statement: The pre-source rank-four relation is tested against complete source-bound galaxy rows and residuals.

Forward derivation

Carrier: galaxy-luminosity-velocity-carrier. **Relation:** rank-four-dilution-scaling-test. **Retained record:** band-distance-inclination-baryonic-choice-and-residuals-held. **Evidence boundary:** sample-and-regression-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-FEEDBACK-001, SFT-ASTRO-GALACTIC-DYNAMICS-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

galaxy-luminosity-velocity-carrier__sample-and-regression-bounded__rank-four-dilution-scaling-test__band-distance-inclination-baryonic-choice-and-residuals-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if regression choice, selection or adverse rows are erased.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 1; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- SPARC-BTFR-001 - transport captured; snapshot
sha256:a9b5776f6812398abcec32c21c6c53e5d0bd88c22a708c9f875e49bd12a3a66f.

Quantitative record. The first preregistered all-row unweighted slope was

3.4863446052195514734204060355305148877413196602651; its deterministic delete-one range was

3.4662293923947706653191129549129334941032620165646 to

3.5477278299570478055812917404758994609387218522174, excluding four. Status:

adverse_first_protocol_preserved. It remains adverse. The separate source-authored method reports 3.85 ± 0.09 , with a stated systematic interval 3.5–4.0; exact four lies at that boundary. Addendum blind status:

not_blind_measurement_already_known_after_first_adverse. No first-result reclassification occurred.

What it means

This law makes **luminosity-rotation scaling** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:8fc3fa297ef292c50c016a29b21345e5598dfe499a04b6bc7c40ab19d9520cf1.

Independent certificate:

sha256:33860819d5a160017a0897edadbd4bcd725cde39e624fd82a798ba49eb5065a9. Engine receipt:

sha256:c3828b9cd507fa3d9b9abf663f4aefd6b4348b414d424262b843d05a45aa295e. Candidate count:

256. Survivor count: 1.

41. Galaxy evolution

Claim: SFT-ASTRO-GALAXY-EVOLUTION-001

Family: galaxies_populations

Statement: Galaxy evolution distinguishes population sequence from individual causal history.

Forward derivation

Carrier: time-ordered-galaxy-population-carrier. Relation: assembly-transformation-quenching-relation. Retained record: selection-redshift-mass-environment-and-progenitors-held. Evidence boundary: population-history-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-FEEDBACK-001, SFT-ASTRO-TULLY-FISHER-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
time-ordered-galaxy-population-carrier__population-history-bounded__assembly-transformation-quenching-relation__selection-redshift-mass-environment-and-progenitors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retain-s-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject progenitor bias or redshift selection erasure.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.

- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.
- NASA-IRSA-001 - transport captured; snapshot
sha256:1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.

What it means

This law makes **galaxy evolution** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:8e5d877919a8e2a30ddf2314f51739628ff7874a5457d462761a2d895252c1d8.

Independent certificate:

sha256:f361fc5347efef4d11d54551aff54dcbad7d22433f97873589a7140db8fe6c9a. Engine receipt:
sha256:b8cdebfea3ab8986671702da489624ec8c20478bae24ce8bcfe4841aa4e2dd98. Candidate count:
256. Survivor count: 1.

42. Groups, clusters and environment

Claim: SFT-ASTRO-CLUSTER-LARGE-ENVIRONMENT-001

Family: galaxies_populations

Statement: Cluster inference retains membership, gas, potential, selection and mass method.

Forward derivation

Carrier: member-galaxy-gas-potential-carrier. Relation:
membership-binding-environment-relation. Retained record:
selection-redshift-membership-mass-method-and-errors-held. Evidence boundary:
cluster-definition-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-FEEDBACK-001, SFT-ASTRO-GALAXY-EVOLUTION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

member-galaxy-gas-potential-carrier__cluster-definition-bounded__membership-binding-e
nvironment-relation__selection-redshift-membership-mass-method-and-errors-held__evid

nce-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if projection or one mass proxy is treated as exact membership and mass.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.
- NASA-IRSA-001 - transport captured; snapshot
sha256:1521537fa6777d5f0e51e8fc7743a73ebc98dd653405032ec51b17ede7045ca1.

What it means

This law makes **groups, clusters and environment** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:5f2062148b5b5b2f7348684026e17e8927ec35ae8207c2642a57b75e0796db71.

Independent certificate:

sha256:4fec8165d72ef8fe565f90d7ecb3524da369103dea25d5643458ac2ed160aaf. Engine receipt:
sha256:83121167316eb91e192b2ecaabdfd73cee80c7db739a5cc1d3d0267f1c288483. Candidate count:
256. Survivor count: 1.

43. Compact-object identity

Claim: SFT-ASTRO-COMPACT-OBJECT-001

Family: compact_high_energy_multimessenger

Statement: Compact-object identity retains source, environment, signals, method and alternatives.

Forward derivation

Carrier: source-field-matter-signal-carrier. **Relation:** compact-state-to-observable-relation. **Retained record:** position-mass-spin-environment-method-and-errors-held. **Evidence boundary:** compact-inference-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,

SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
 SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
 SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
 SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-CLUSTER-LARGE-ENVIRONMENT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
source-field-matter-signal-carrier__compact-inference-bounded__compact-state-to-observable-relation__position-mass-spin-environment-method-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if one signature uniquely fixes an unobserved interior.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256:66771560db1200bc22064d0b7cbe2515a98292d28f6e664f79beaf43d8ba0382.
- ICECUBE-PUBLIC-DATA-001 - transport captured; snapshot
sha256:8f2fb1ac5bc28957f49b097d2a17bad60452b06b3302f395afdda597db29535e.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **compact-object identity** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:5c4f0f1bd4eb68485999d5fd6c8b5235fc8a6f473f7628b2b84e8855a971de58.

Independent certificate:

sha256:008d9d8ca098986bb1731c78ae7f6f6b5d10a59648f4e42c22dac58de8b65568. Engine receipt:

sha256:1d72a9707b68e99d69093885946b890ebdb3d08dffdb8e1a9c0d166335c90ec6. Candidate count:

256. Survivor count: 1.

44. Degenerate remnant classes

Claim: SFT-ASTRO-WHITE-DWARF-NEUTRON-STAR-001

Family: compact_high_energy_multimessenger

Statement: White-dwarf and neutron-star claims retain mass, radius, composition, rotation and field evidence.

Forward derivation

Carrier: progenitor-degenerate-remnant-carrier. Relation: mass-composition-pressure-endpoint-relation. Retained record: mass-radius-composition-field-rotation-and-errors-held. Evidence boundary: remnant-class-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-CLUSTER-LARGE-ENVIRONMENT-001, SFT-ASTRO-COMPACT-OBJECT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

progenitor-degenerate-remnant-carrier__remnant-class-bounded__mass-composition-pressure-endpoint-relation__mass-radius-composition-field-rotation-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least

one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if a theoretical limit replaces measurement.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256:66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.
- ICECUBE-PUBLIC-DATA-001 - transport captured; snapshot
sha256:8f2fb1ac5bc28957f49b097d2a17bad60452b06b3302f395afdda597db29535e.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **degenerate remnant classes** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:4595030afd53e4ebc85ce9cf2cc76c3479f83f34be28b50c5a5ad78d2c5b37d3.

Independent certificate:

sha256:3915e446e6a6828ef287771d2ad41b813f228a4166affd68f1dd0fc3bf0b9259. Engine receipt:
sha256:8270a0c8773e17094236ae656d4872538b39c7c3153f3c9149a7cba014896dd0. Candidate count:
256. Survivor count: 1.

45. Black-hole candidate and horizon evidence

Claim: SFT-ASTRO-BLACK-HOLE-001

Family: `compact_high_energy_multimessenger`

Statement: A black-hole claim retains dynamical mass, compactness, environment, messenger and alternatives.

Forward derivation

Carrier: `source-orbit-accretion-wave-carrier`. Relation:
`compactness-and-horizon-consequence-relation`. Retained record:
`mass-spin-environment-messenger-and-alternatives-held`. Evidence boundary:
`horizon-evidence-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,

SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-CLUSTER-LARGE-ENVIRONMENT-001, SFT-ASTRO-WHITE-DWARF-NEUTRON-STAR-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
source-orbit-accretion-wave-carrier__horizon-evidence-bounded__compactness-and-horizo
n-consequence-relation__mass-spin-environment-messenger-and-alternatives-held__eviden
ce-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-
rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if darkness alone proves a horizon or interior ontology.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256:66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.
- ICECUBE-PUBLIC-DATA-001 - transport captured; snapshot
sha256:8f2fblac5bc28957f49b097d2a17bad60452b06b3302f395afdda597db29535e.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **black-hole candidate and horizon evidence** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:75f47ad8561ae08e46cf6a708882b91ae28b417818c52281154e58c1551a2872.

Independent certificate:

sha256:e56f7991938117720cfe79f8951b5fe70ef3729115c2d3bb0f7b3e8ae44c9e44. Engine receipt: sha256:7764fe13a9fb7b3d43062fec060acf9379bf00321f0bbcab8439813fbf3ff9ec. Candidate count: 256. Survivor count: 1.

46. High-energy transient

Claim: SFT-ASTRO-TRANSIENT-001

Family: compact_high_energy_multimessenger

Statement: A transient links source, event, localization, time, bands and afterglow with alternatives.

Forward derivation

Carrier: source-event-multiband-carrier. **Relation:** progenitor-to-transient-relation. **Retained record:** localization-time-bands-selection-afterglow-and-errors-held. **Evidence boundary:** transient-class-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-CLUSTER-LARGE-ENVIRONMENT-001, SFT-ASTRO-BLACK-HOLE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
source-event-multiband-carrier__transient-class-bounded__progenitor-to-transient-relation__localization-time-bands-selection-afterglow-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject if one light-curve feature fixes a unique progenitor.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256:66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.
- ICECUBE-PUBLIC-DATA-001 - transport captured; snapshot
sha256:8f2fb1ac5bc28957f49b097d2a17bad60452b06b3302f395afdda597db29535e.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **high-energy transient** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:0694ae0d2d9438317d4d7e282e233201992c84d8ad33d70209536cb435e9c342.

Independent certificate:

sha256:ea4bc7298122faff2a7aa03f8968a813e71bf93e5577edcef4e3d1d1da5c1b50. Engine receipt:
sha256:c5eac33d788970c7c587923786815268fcb4c635f2f2469e16d08dc378136291. Candidate count:
256. Survivor count: 1.

47. Multi-messenger association

Claim: SFT-ASTRO-MULTIMESSENGER-001

Family: `compact_high_energy_multimessenger`

Statement: Multi-messenger association retains localization, clocks, selection and background coincidence.

Forward derivation

Carrier: `photon-wave-neutrino-particle-event-carrier`. Relation:
`space-time-source-association-relation`. Retained record:
`localizations-clocks-selection-background-and-errors-held`. Evidence boundary:
`association-window-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,

SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-CLUSTER-LARGE-ENVIRONMENT-001, SFT-ASTRO-TRANSIENT-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

photon-wave-neutrino-particle-event-carrier__association-window-bounded__space-time-s
 ource-association-relation__localizations-clocks-selection-background-and-errors-held
 __evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-reta
 ins-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if temporal proximity alone proves common origin.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement /
 primary_archive_or_measurement_correspondence. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- GWOSC-PUBLIC-001 - transport captured; snapshot
 sha256:66771560db1200bc22064d0b7cbe2515a98292d28f6e664f79beaf43d8ba0382.
- ICECUBE-PUBLIC-DATA-001 - transport captured; snapshot
 sha256:8f2fb1ac5bc28957f49b097d2a17bad60452b06b3302f395afdda597db29535e.
- NASA-HEASARC-001 - transport captured; snapshot
 sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **multi-messenger association** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:f473826f9c3736a49058496d045ec507c3621eb84b9d0b7da72b18c1b155ccc7.

Independent certificate:

sha256:b974c2ce6fd98dd5f52076b68657d41e325d9059474d16d38f24fa3554dc7b88. Engine receipt:
 sha256:98f412670daf803b95eadc7246a7a5749e56caaf065a2266f008cd4be971e921. Candidate count:
 256. Survivor count: 1.

48. Compact-object population

Claim: SFT-ASTRO-COMPACT-POPULATION-001

Family: compact_high_energy_multimessenger

Statement: Compact populations retain detection selection, censoring and source history.

Forward derivation

Carrier: selection-bounded-remnant-set-carrier. **Relation:** formation-merger-population-relation. **Retained record:** survey-volume-selection-masses-spins-and-censoring-held. **Evidence boundary:** population-model-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-CLUSTER-LARGE-ENVIRONMENT-001, SFT-ASTRO-MULTIMESSENGER-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
selection-bounded-remnant-set-carrier__population-model-bounded__formation-merger-population-relation__survey-volume-selection-masses-spins-and-censoring-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject if detected events are treated as an unbiased population.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 3; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256:66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.
- ICECUBE-PUBLIC-DATA-001 - transport captured; snapshot
sha256:8f2fb1ac5bc28957f49b097d2a17bad60452b06b3302f395afdda597db29535e.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **compact-object population** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:3d09c6675f037b5b71b07caf52e448b9ba1617c4e33939c735641641b171de8a.

Independent certificate:

sha256:36b94990d08eedf60dbeff556d0b2b5b41651291473f75f020847a9b52ca4aa6. Engine receipt:
sha256:6b6de3d5331b1c9e197bada2a99f2fb8c4685f13c83714b7d0d178a7ecf6c17d. Candidate count:
256. Survivor count: 1.

49. Large-scale statistical organisation

Claim: SFT-ASTRO-COSMIC-HOMOGENEITY-001

Family: `large_scale_expansion`

Statement: Large-scale organization is a survey- and scale-bound tracer distribution.

Forward derivation

Carrier: `survey-volume-tracer-field-carrier`. Relation: `scale-dependent-distribution-relation`. Retained record: `selection-mask-redshift-distance-and-errors-held`. Evidence boundary: `survey-scale-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-COMPACT-POPULATION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
survey-volume-tracer-field-carrier__survey-scale-bounded__scale-dependent-distributio
n-relation__selection-mask-redshift-distance-and-errors-held__evidence-class-explicit
__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-ru
le
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if finite-volume uniformity is asserted without scale and selection.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **large-scale statistical organisation** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:4046956aca5d95c6e5eea080d9c99b39412c6affdca6d0d01014d15e2ca74a2e.

Independent certificate:

sha256:6c7283928481632b0830011a78b1721743780d4a882ee175fc8ccb24d8f266ce. Engine receipt:
sha256:7e928a24126ea2671b1fddb0fd3d38e85e0fece770f5d579573517f9205b18ea. Candidate count:
256. Survivor count: 1.

50. Cosmic-web structure

Claim: SFT-ASTRO-COSMIC-WEB-001

Family: large_scale_expansion

Statement: Web structure retains tracer, scale, method, selection and redshift.

Forward derivation

Carrier: tracer-node-filament-void-carrier. **Relation:** adjacency-density-history-relation. **Retained record:** tracer-selection-scale-redshift-method-and-errors-held. **Evidence boundary:** web-extraction-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-COMPACT-POPULATION-001, SFT-ASTRO-COSMIC-HOMOGENEITY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
tracer-node-filament-void-carrier__web-extraction-bounded__adjacency-density-history-
relation__tracer-selection-scale-redshift-method-and-errors-held__evidence-class-expl
icit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extr
a-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject if a visualization substitutes for a generated structure criterion.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. **Captured sources:** 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. **External evidence selected the survivor:** false.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **cosmic-web structure** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:b450b3498e6501f5683585ebb05f460e0cd345416fd8b93b2769e7a8df62bfdc.

Independent certificate:

sha256:551f3db8e5a3aa0bcfcf1b8c46c4a038029308de462545f8c2b3b9144f0c994f. Engine receipt:
sha256:473d2d8bf8837e40ef8530f0aedd56aff5cfe10a3f0f59b146be1a0ea85711b. Candidate count:
256. Survivor count: 1.

51. Cosmic expansion record

Claim: SFT-ASTRO-EXPANSION-RECORD-001

Family: large_scale_expansion

Statement: Expansion is a multi-probe distance-redshift record with calibration and covariance retained.

Forward derivation

Carrier: source-redshift-distance-time-carrier. Relation:
distance-redshift-comparison-relation. Retained record:
standardization-calibration-selection-covariance-and-era-held. Evidence boundary:
expansion-probe-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-COMPACT-POPULATION-001, SFT-ASTRO-COSMIC-WEB-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

source-redshift-distance-time-carrier__expansion-probe-bounded__distance-redshift-com
parison-relation__standardization-calibration-selection-covariance-and-era-held__evid

ence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if one rung or probe silently sets the law.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **cosmic expansion record** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:2726011cfa7cd209bcb43d2179aada0cf289cc6a6834d7c45dec50e9fa2a0eba.

Independent certificate:

sha256:535c0cf9449089358a0c5656397adc601c886bb69986fca502396fb49bc2a94e. Engine receipt:

sha256:d08db815fc6f99661fd7e78f5f67a40e4734053c672d5cb5d053bde2f349bd75. Candidate count:

256. Survivor count: 1.

52. Distance-ladder composition

Claim: SFT-ASTRO-DISTANCE-LADDER-001

Family: large_scale_expansion

Statement: A distance ladder is an explicit composition of anchors and rungs with covariance.

Forward derivation

Carrier: anchor-rung-source-carrier. Relation: calibration-composition-relation. Retained record: anchors-rungs-populations-zero-points-and-covariance-held. Evidence boundary: ladder-version-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,

SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
 SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
 SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
 SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-COMPACT-POPULATION-001, SFT-ASTRO-EXPANSION-RECORD-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
anchor-rung-source-carrier__ladder-version-bounded__calibration-composition-relation__
anchors-rungs-populations-zero-points-and-covariance-held__evidence-class-explicit__
root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a terminal value is detached from its ladder.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
 sha256: 7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- SDSS-DATA-001 - transport captured; snapshot
 sha256: 67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **distance-ladder composition** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:4310d44d9fdd0b3bc9308301455dab439637b5f2cd34b3c84a8d5f4c32444ae4.

Independent certificate:

sha256:86f1e2316b08c0b14890579c7d2d0c41c987f1690c05bfe9c946c27f1e944abe. Engine receipt: sha256:09bb53ce938c971978590248d20c8241f162a071f7a820fa660cfebb82d2139c. Candidate count: 256. Survivor count: 1.

53. Gravitational-lensing observation

Claim: SFT-ASTRO-LENSING-001

Family: large_scale_expansion

Statement: Lensing retains source, lens, observer, geometry, mass model and degeneracies.

Forward derivation

Carrier: source-lens-observer-image-carrier. Relation: multiple-path-distortion-relation.

Retained record: redshifts-geometry-selection-mass-model-and-errors-held. Evidence boundary: lensing-model-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-COMPACT-POPULATION-001, SFT-ASTRO-DISTANCE-LADDER-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
source-lens-observer-image-carrier__lensing-model-bounded__multiple-path-distortion-r
elation__redshifts-geometry-selection-mass-model-and-errors-held__evidence-class-expl
icit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extr
a-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if an image distortion uniquely identifies mass without alternatives.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **gravitational-lensing observation** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:47285b2539d8c9304fa82644849177e726007aa2fda09074468313318d144b39.

Independent certificate:

sha256:dfecb487549ac37f21ea490b03bea477467ebff9d97a51b0b33d564ffb5a192d. Engine receipt:

sha256:aa79bb09787cd670d87fd4a7ddfadf08bba448490936a246b6e7600d46bac534. Candidate count: 256. Survivor count: 1.

54. Expansion-history inference

Claim: SFT-ASTRO-EXPANSION-HISTORY-001

Family: `large_scale_expansion`

Statement: Expansion history is an ordered reconstruction from independently retained probes.

Forward derivation

Carrier: `multi-probe-epoch-carrier`. **Relation:** `ordered-expansion-state-reconstruction`. **Retained record:** `probes-calibration-covariance-model-boundary-and-time-held`. **Evidence boundary:** `historical-inference-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,

SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-COMPACT-POPULATION-001, SFT-ASTRO-LENSING-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
multi-probe-epoch-carrier__historical-inference-bounded__ordered-expansion-state-reco
nstruction__probes-calibration-covariance-model-boundary-and-time-held__evidence-clas
s-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__n
o-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a fitted curve is relabelled direct cosmic history.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement /
primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- SDSS-DATA-001 - transport captured; snapshot
sha256:67e7ad58efa8a68f44ea7b1b948659e6d7e1d5538558a9dcea85138e38f29c0e.

What it means

This law makes **expansion-history inference** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:809205f910e2bd1deacda890b50684ba5728975c3e681d4c2791fe14a621c225.

Independent certificate:

sha256:16f1fc4b89e3c87996e766c7769129fb10431fe7c567224f3a1eac31e43c07dd. Engine receipt:
sha256:f49381be08c9ea93843dd3b65e22c9cd26a9fbba329cd6c3ab085d55503b5d36. Candidate count:
256. Survivor count: 1.

55. Cosmic thermal history

Claim: SFT-ASTRO-THERMAL-HISTORY-001

Family: early_background_abundance

Statement: Thermal history retains species, interaction, scale, temperature and evidence class by epoch.

Forward derivation

Carrier: ordered-radiation-matter-state-carrier. **Relation:** dilution-interaction-transition-relation. **Retained record:** temperature-scale-species-epoch-and-evidence-held. **Evidence boundary:** thermal-reconstruction-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-EXPANSION-HISTORY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
ordered-radiation-matter-state-carrier__thermal-reconstruction-bounded__dilution-interaction-transition-relation__temperature-scale-species-epoch-and-evidence-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject if a present background uniquely supplies all earlier states.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **cosmic thermal history** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:a62a3d56c17d8401ae5711c63d3b4765fc7549bf9686c81aebf1889bd3502738.

Independent certificate:

sha256:a14dcfd5e9ac37c8e334522382bbcb1e83cd6791740cfa12cbbce5a357f011db. Engine receipt:
sha256:c2f4e22ae5b613c94534375ee96900a2751ff1cfc8f8ea35c3ae74c004907018. Candidate count:
256. Survivor count: 1.

56. Cosmic microwave background

Claim: SFT-ASTRO-CMB-001

Family: `early_background_abundance`

Statement: CMB evidence retains frequency, calibration, beam, masks, foregrounds and covariance.

Forward derivation

Carrier: `all-sky-frequency-temperature-polarization-carrier`. Relation: `background-anisotropy-spectrum-relation`. Retained record: `frequency-calibration-mask-beam-foregrounds-and-covariance-held`. Evidence boundary: `cmb-release-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-EXPANSION-HISTORY-001, SFT-ASTRO-THERMAL-HISTORY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
all-sky-frequency-temperature-polarization-carrier__cmb-release-bounded__background-a
nisotropy-spectrum-relation__frequency-calibration-mask-beam-foregrounds-and-covarian
ce-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extensi
on-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a component-separated map is called raw observation.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **cosmic microwave background** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:806c1220714f451e2fc205792b13182a2a524e003b0026cce6d8e82ddb9d1e47.

Independent certificate:

sha256:f7b966300db8bc75a9a123f7e60c4045aa9db65bff4777597df852a834c321af. Engine receipt:
sha256:7118549efc4e4f62f0a3f11bd35f87cc36682d2f557515aeea40d8b0b3d5a557. Candidate count:
256. Survivor count: 1.

57. Background acoustic structure

Claim: SFT-ASTRO-ACOUSTIC-STRUCTURE-001

Family: early_background_abundance

Statement: Acoustic structure retains harmonic features, covariance and extraction boundary.

Forward derivation

Carrier: angular-spectrum-feature-carrier. **Relation:** bounded-harmonic-recurrence-relation. **Retained record:** multipoles-peaks-covariance-foregrounds-and-model-boundary-held. **Evidence boundary:** spectrum-analysis-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-EXPANSION-HISTORY-001, SFT-ASTRO-CMB-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
angular-spectrum-feature-carrier__spectrum-analysis-bounded__bounded-harmonic-recurrence-relation__multipoles-peaks-covariance-foregrounds-and-model-boundary-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject if a selected peak or harmonic fit erases adverse multipoles.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. **Captured sources:** 2; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.

- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **background acoustic structure** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:929c153b2747f4c46f0795820af65f787e2489853adf87460a47e58a8dc13660.

Independent certificate:

sha256:f2657b826511392ac1acb8561885144d38e97057aefc23c0276a2c71e732749d. Engine receipt:

sha256:897cd20cee5b797d7833de676b37dbd87a27ad30282b26fc64357198366bbea4. Candidate count:

256. Survivor count: 1.

58. Primordial abundance record

Claim: SFT-ASTRO-PRIMORDIAL-ABUNDANCE-001

Family: early_background_abundance

Statement: Primordial abundance inference retains species, sites, methods, depletion and stellar processing.

Forward derivation

Carrier: species-environment-abundance-carrier. **Relation:** reaction-history-to-abundance-relation. **Retained record:** species-sites-methods-depletion-and-errors-held. **Evidence boundary:** abundance-inference-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-EXPANSION-HISTORY-001, SFT-ASTRO-ACOUSTIC-STRUCTURE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

species-environment-abundance-carrier__abundance-inference-bounded__reaction-history-to-abundance-relation__species-sites-methods-depletion-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if current abundance is silently relabelled primordial.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **primordial abundance record** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:de4937f016e57a3b955b8eed89ed8e61f3b53a3365954fc1aaab4557e19aba5c.

Independent certificate:

sha256:3025487a0b98dc72a526d2c12080f7db19a78f5b6429e320f50ac4997579a654. Engine receipt:

sha256:7cf55eed144d2fad2357efe5c54fdc92510151d15e73c07321723395df4883fb. Candidate count: 256. Survivor count: 1.

59. Recombination and photon release

Claim: SFT-ASTRO-RECOMBINATION-001

Family: `early_background_abundance`

Statement: Recombination links matter state, opacity and photon release with evidence boundaries.

Forward derivation

Carrier: `matter-radiation-interaction-state-carrier`. Relation:

`binding-opacity-release-relation`. Retained record:

`species-temperature-redshift-optical-depth-and-evidence-held`. Evidence boundary:

`recombination-model-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,

SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-EXPANSION-HISTORY-001, SFT-ASTRO-PRIMORDIAL-ABUNDANCE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
matter-radiation-interaction-state-carrier__recombination-model-bounded__binding-opac
ity-release-relation__species-temperature-redshift-optical-depth-and-evidence-held__e
vidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains
-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if a model epoch is relabelled direct observation.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-LAMBDA-001 - transport captured; snapshot
 sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- NASA-HEASARC-001 - transport captured; snapshot
 sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **recombination and photon release** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:732c4d0c5825a9b1afba6ebac6b66396893a90dcfef87a002ecc1e4856d74bc9.

Independent certificate:

sha256:0d1fdbce9bcd44fdd39dac1d14277979597e9c685886c0ca2483b7d36b0acd0. Engine receipt:
sha256:55693de3cc33c46c54235ea2ed160fb034a7bf81a5af2ce2ad5a6c5f7d709843. Candidate count:
256. Survivor count: 1.

60. Early perturbation record

Claim: SFT-ASTRO-INITIAL-PERTURBATION-001

Family: early_background_abundance

Statement: An early perturbation record retains scale, spectrum, covariance and transfer to later tracers.

Forward derivation

Carrier: background-anisotropy-seed-carrier. Relation:
initial-distinction-to-later-structure-relation. Retained record:
scale-spectrum-phase-covariance-and-transfer-held. Evidence boundary:
seed-inference-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-EXPANSION-HISTORY-001, SFT-ASTRO-RECOMBINATION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

background-anisotropy-seed-carrier__seed-inference-bounded__initial-distinction-to-later-structure-relation__scale-spectrum-phase-covariance-and-transfer-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if later structure uniquely reconstructs an unrecorded initial state.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 0. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- NASA-HEASARC-001 - transport captured; snapshot
sha256:030935b4594735c55734bb14156d84cfe67a0946f8c19158aae7101b8382edfe.

What it means

This law makes **early perturbation record** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:d8e02965bdf73aa4bc70a8ef1603beeeb4b5f6d66b98b6044c7a2e0c5f490238.

Independent certificate:

sha256:abed46a2470051f839e8ddd1bb8aa85914f6eacfdccc77250ed2a7f107bb4d7ea. Engine receipt: sha256:2cfecdf1565cc5f5bcf85154beb1e17120cbe8188b01418c605e77934803ac2f. Candidate count: 256. Survivor count: 1.

61. Dark-matter evidence class

Claim: SFT-ASTRO-DARK-MATTER-EVIDENCE-001

Family: `dark_horizon_unobserved`

Statement: Dark-matter evidence is a cross-probe residual with baryons and alternatives retained.

Forward derivation

Carrier: `visible-tracer-gravitational-residual-carrier`. Relation: `observed-motion-lensing-growth-comparison`. Retained record: `baryons-geometry-selection-alternatives-and-errors-held`. Evidence boundary: `dark-inference-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,

SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-INITIAL-PERTURBATION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
visible-tracer-gravitational-residual-carrier__dark-inference-bounded__observed-motion-lensing-growth-comparison__baryons-geometry-selection-alternatives-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if one anomaly or prior preference uniquely selects a carrier.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256:66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.

What it means

This law makes **dark-matter evidence class** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:2489c6ef560a7807256e6bc9797a24c95adb8e9a0cd2dde0e1e0d748d0cbe698.

Independent certificate:

sha256:add825a43217ee256a5214aaf2d6443b2d069d38ca9064a2d0a999a2f96abee. Engine receipt:
sha256:da36b528cee229d263a4537a6886b66fb0c5e7448bf0f8d08e32158f31945981. Candidate count:
256. Survivor count: 1.

62. Dark-energy evidence class

Claim: SFT-ASTRO-DARK-ENERGY-EVIDENCE-001

Family: dark_horizon_unobserved

Statement: Dark-energy evidence retains distance, growth, calibration, covariance and alternatives.

Forward derivation

Carrier: distance-growth-expansion-residual-carrier. **Relation:** multi-probe-acceleration-comparison. **Retained record:** calibration-covariance-model-boundary-and-alternatives-held. **Evidence boundary:** dark-energy-inference-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-INITIAL-PERTURBATION-001, SFT-ASTRO-DARK-MATTER-EVIDENCE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
distance-growth-expansion-residual-carrier__dark-energy-inference-bounded__multi-prob
e-acceleration-comparison__calibration-covariance-model-boundary-and-alternatives-hel
d__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-ret
ains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject if acceleration data alone fixes ontology or an equation of state.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256:66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.

What it means

This law makes **dark-energy evidence class** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:51b09808f74a77e4c89c9e499ef64d5a495646bd37de06c75657990b74c91b92.

Independent certificate:

sha256:a951e538d6f141760d4c5050ee9281147638ab45fdaa855a3b214eb333a2f716. Engine receipt:
sha256:c65dbbb54d1c96f9c2c02534f42eac5c4523f3c8ef09256f33013b3764437f8f. Candidate count:
256. Survivor count: 1.

63. Cosmic horizon

Claim: SFT-ASTRO-COSMIC-HORIZON-001

Family: `dark_horizon_unobserved`

Statement: A horizon is an observer- and epoch-bound causal access limit under admitted propagation.

Forward derivation

Carrier: `observer-signal-reach-carrier`. Relation: `finite-history-causal-access-relation`.

Retained record: `observer-epoch-propagation-law-and-boundary-held`. Evidence boundary:

`horizon-definition-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-INITIAL-PERTURBATION-001, SFT-ASTRO-DARK-ENERGY-EVIDENCE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
observer-signal-reach-carrier__horizon-definition-bounded__finite-history-causal-access-relation__observer-epoch-propagation-law-and-boundary-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if unobserved beyond-horizon state is relabelled observed absence.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256:66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.

What it means

This law makes **cosmic horizon** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:23c9f22f9583241e69a07c3ec8f611fb473a2ddd8e30194ca2fd476a6bd0c82d.

Independent certificate:

sha256:fe1e7e05abf7e5550490f8845f4bc5893b1f25ff003effd155dfdc248308172. Engine receipt:
sha256:973035e63119a5e728bb88f8606b633621a1f28d5c3a63a50e925504c79e42cd. Candidate count:
256. Survivor count: 1.

64. Cosmic topology claim

Claim: SFT-ASTRO-TOPOLOGY-001

Family: dark_horizon_unobserved

Statement: Topology claims require observable repeated-pattern consequences and controls.

Forward derivation

Carrier: observable-pattern-identification-carrier. **Relation:** global-identification-consequence-relation. **Retained record:** survey-mask-scale-pattern-controls-and-alternatives-held. **Evidence boundary:** observable-domain-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-INITIAL-PERTURBATION-001, SFT-ASTRO-COSMIC-HORIZON-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
observable-pattern-identification-carrier__observable-domain-bounded__global-identification-consequence-relation__survey-mask-scale-pattern-controls-and-alternatives-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject if mathematical possibility is counted as observation.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. **Captured sources:** 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.

- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256:66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.

What it means

This law makes **cosmic topology claim** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:c187945ecee49a6ec909672e915d4df55cd28dd6b7e00f33b08bfe7dc198d342.

Independent certificate:

sha256:88d880d117bd5d5fde68df9e2c105680bc617dfd56c32121da5869f1b8bb3ff1. Engine receipt:
sha256:0040739a2ddeb6c99077b83d3fe789f427c38f159274e8f9495274ee1b7fa0e7. Candidate count:
256. Survivor count: 1.

65. Multiverse admissibility boundary

Claim: SFT-ASTRO-MULTIVERSE-001

Family: dark_horizon_unobserved

Statement: A multiverse claim enters empirical science only through a distinct observable consequence sealed before data.

Forward derivation

Carrier: declared-domain-hypothesis-carrier. Relation:
observable-consequence-or-nonadmission-relation. Retained record:
premises-domain-predictions-controls-and-access-held. Evidence boundary:
observable-consequence-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,
SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
SFT-ASTRO-INITIAL-PERTURBATION-001, SFT-ASTRO-TOPOLOGY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

declared-domain-hypothesis-carrier__observable-consequence-bounded__observable-consequence-or-nonadmission-relation__premises-domain-predictions-controls-and-access-held__

`__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule`

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if unobservability, anthropic selection or possibility is treated as confirmation.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- GWOSC-PUBLIC-001 - transport captured; snapshot
sha256:66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.

What it means

This law makes **multiverse admissibility boundary** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:cb6024d3df8c3f3cdd5558351a5283c8e26100fb231e81bd9e988bc9f41c0db8.

Independent certificate:

sha256:fe07573fb0b37b59a2c944112e25e5570432212c1ecb52b0827cdc69634d413f. Engine receipt:

sha256:a117824bcf01755b83dc742009d2aad4339370fc5bf9c9626a1d3b8c7e494b6c. Candidate count:

256. Survivor count: 1.

66. Cosmic ultimate-fate inference

Claim: SFT-ASTRO-ULTIMATE-FATE-001

Family: `dark_horizon_unobserved`

Statement: Ultimate fate is conditional continuation from current records and admitted laws, not an observed terminal state.

Forward derivation

Carrier: `current-state-law-history-carrier`. Relation:

`conditional-future-continuation-relation`. Retained record:

`state-laws-uncertainty-alternatives-and-domain-held`. Evidence boundary:

`future-inference-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,

SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
 SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001,
 SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001,
 SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001,
 SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,
 SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001,
 SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001,
 SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001,
 SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001,
 SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067,
 SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,
 SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001,
 SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001,
 SFT-ASTRO-INITIAL-PERTURBATION-001, SFT-ASTRO-MULTIVERSE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
current-state-law-history-carrier__future-inference-bounded__conditional-future-continuation-relation__state-laws-uncertainty-alternatives-and-domain-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject if extrapolation uncertainty or alternative lawful continuation is erased.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-LAMBDA-001 - transport captured; snapshot
 sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- DESI-DR1-001 - transport failed_preserved; snapshot unresolved.
- GWOSC-PUBLIC-001 - transport captured; snapshot
 sha256:66771560db1200bc22064d0b7cbe2515a98292d28fce664f79beaf43d8ba0382.

What it means

This law makes **cosmic ultimate-fate inference** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:74b97003c83b3c9ac71d4b1228378463906607d074c568859bd5ac52204f73eb.

Independent certificate:

sha256:7719f41c318f8ac503948dc16e47e28663521c1954a3d6e1c20f61453e704a87. Engine receipt:

sha256:49730cfa0cd10c71cf80dd286733833e51dee356deca785b1fa0ee10e82867f6. Candidate count:

256. Survivor count: 1.

67. Universal law and cosmic history distinction

Claim: SFT-ASTRO-LAW-HISTORY-001

Family: inference_prediction_handoffs

Statement: Universal law, contingent initial condition and source-dated history remain separate.

Forward derivation

Carrier: law-initial-condition-observation-carrier. Relation: lawful-history-generation-relation. Retained record: law-receipts-initial-records-and-observations-held. Evidence boundary: ownership-boundary-declared.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ULTIMATE-FATE-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
law-initial-condition-observation-carrier__ownership-boundary-declared__lawful-history-generation-relation__law-receipts-initial-records-and-observations-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least

one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if a catalogue regularity is promoted to premise-free law.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- NASA-ADS-001 - transport failed; preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.

What it means

This law makes **universal law and cosmic history distinction** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:7615b676bcb1a70d00b85d9e6451ee6b15cecfde0efdc03c848bee376d75fe78.

Independent certificate:

sha256:c7410f1f7c30b2f43c2abe6ac06c3468d2992b359072531fe29f36e5e5d00736. Engine receipt:
sha256:c3a3d30c99d99d46c7a26df839bec5d9cd41351cd48e5ab40c5ed9cb0bb58796. Candidate count:
256. Survivor count: 1.

68. Astronomical model-observation boundary

Claim: SFT-ASTRO-MODEL-OBSERVATION-001

Family: `inference_prediction_handoffs`

Statement: A model output remains a conditional result compared with, not substituted for, observation.

Forward derivation

Carrier: `law-state-instrument-model-output-carrier`. Relation: `conditional-simulation-comparison-relation`. Retained record: `inputs-parameters-observations-errors-and-output-held`. Evidence boundary: `model-purpose-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001,

SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ULTIMATE-FATE-001, SFT-ASTRO-LAW-HISTORY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
law-state-instrument-model-output-carrier__model-purpose-bounded__conditional-simulation-comparison-relation__inputs-parameters-observations-errors-and-output-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject opaque prediction or parameter fitting as derivation.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-ADS-001 - transport failed_preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.

What it means

This law makes **astronomical model-observation boundary** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:d0cc6db940af599f3d199b2ec899c3d45940fabdf814ca78afbb9430cfce5a44.

Independent certificate:

sha256:f7a47e8727ea6a1d74056fbc7f882bcc8b617d9048b2e7fa8741a703c5ac61cc. Engine receipt: sha256:887f1ed59a17938a9603af5790b6e13dbcb1c63cdd768242a0a89dcfb19b8991. Candidate count: 256. Survivor count: 1.

69. Standing astronomical prediction

Claim: SFT-ASTRO-STANDING-PREDICTION-001

Family: inference_prediction_handoffs

Statement: A standing prediction is sealed before outcome and retains every adverse and missing row.

Forward derivation

Carrier: sealed-premise-prediction-outcome-carrier. **Relation:** predeclared-comparison-relation. **Retained record:** targets-metrics-baselines-adverse-and-missing-rows-held. **Evidence boundary:** prediction-question-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ULTIMATE-FATE-001, SFT-ASTRO-MODEL-OBSERVATION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
sealed-premise-prediction-outcome-carrier__prediction-question-bounded__predeclared-comparison-relation__targets-metrics-baselines-adverse-and-missing-rows-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject post-outcome target changes or favourable-row selection.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-ADS-001 - transport failed_preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.

What it means

This law makes **standing astronomical prediction** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:270b6510c1fc80f7274e0323530b0e1176f983d57cfc20812445548ccc6c3f0f.

Independent certificate:

sha256:8b7b81836bef277ed91a661cd5dfab965e26cef89cda75794e5914a0c501de43. Engine receipt:

sha256:b83b752df3a1e802277ac2e5eaeef8205baeefbd00d827236838e8a57ccaa1b93. Candidate count:

256. Survivor count: 1.

70. Unexplained astronomical observation

Claim: SFT-ASTRO-UNEXPLAINED-001

Family: `inference_prediction_handoffs`

Statement: Unexplained means a source-bound residual after declared controls, not evidence for a preferred unknown cause.

Forward derivation

Carrier: `observation-controls-alternative-carrier`. Relation: `residual-after-tested-explanations-relation`. Retained record: `data-method-controls-alternatives-and-failures-held`. Evidence boundary: `residual-evidence-bounded`.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069,

SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ULTIMATE-FATE-001, SFT-ASTRO-STANDING-PREDICTION-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
observation-controls-alternative-carrier__residual-evidence-bounded__residual-after-t
ested-explanations-relation__data-method-controls-alternatives-and-failures-held__evi
dence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-a
ll-rows__no-extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. Minimality: True. Named-form uniqueness: True.

Falsification and controls

Falsification condition: Reject argument from ignorance or hidden adverse controls.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-ADS-001 - transport failed_preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.

What it means

This law makes **unexplained astronomical observation** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:4c5f2f4efe22a0ef5876fe979e0725066510a3eff19d9d55371181f8dbf5a13e.

Independent certificate:

sha256:3d409d8701b71cd241505f4ae9157d5bc38c0d8e4fa2fa8bc5d3ab007651118b. Engine receipt:
sha256:7af6264c33e27e20954db0642d0051f8e9b14af8664c8446daf278ed238bbcf2. Candidate count:
256. Survivor count: 1.

71. Astrobiology and technosignature boundary

Claim: SFT-ASTRO-ASTROBIOLOGY-001

Family: inference_prediction_handoffs

Statement: Life or technology claims require replicated signals and complete abiotic, instrumental and contamination controls.

Forward derivation

Carrier: environment-signal-life-hypothesis-carrier. **Relation:** biosignature-or-technosignature-comparison. **Retained record:** abiotic-controls-selection-contamination-and-replication-held. **Evidence boundary:** life-evidence-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ULTIMATE-FATE-001, SFT-ASTRO-UNEXPLAINED-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

environment-signal-life-hypothesis-carrier__life-evidence-bounded__biosignature-or-technosignature-comparison__abiotic-controls-selection-contamination-and-replication-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-extra-rule

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: depth_independent. **Minimality:** True. **Named-form uniqueness:** True.

Falsification and controls

Falsification condition: Reject an ambiguous signal or habitability proxy as life detection.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: True. External evidence selected the survivor: false.

- NASA-ADS-001 - transport failed_preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.

What it means

This law makes **astrobiology and technosignature boundary** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:1f067d4fba2910dd7062cb29edaa4072315e6ee30c5f6fa2cba36620467f79b0.

Independent certificate:

sha256:ee3c67811dbdb007ae493220f8ab7bc3b1838ad05a1d24ac4168dcbbcb6658451. Engine receipt:
sha256:1b06beceb6d9a84c63a86aef6d02baf535a9244f99301d68a82778fae3d6e585. Candidate count:
256. Survivor count: 1.

72. Astronomy scientific handoffs

Claim: SFT-ASTRO-ASTRO-HANDOFF-001

Family: inference_prediction_handoffs

Statement: Astronomy consumes universal laws and exports source-dated constraints through explicit Physics, Engineering, Earth, Biology and Social boundaries.

Forward derivation

Carrier: law-instrument-earth-life-social-boundary-carrier. Relation: typed-branch-handoff-relation. Retained record: source-owner-dependency-evidence-and-use-held. Evidence boundary: branch-ownership-bounded.

Dependencies: SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-CONSERVATION-LOSS-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-PHYS-MEAS-OBSERVATION-CARRIER-001, SFT-PHYS-MEAS-UNCERTAINTY-001, SFT-PHYS-WAVE-PROPAGATION-001, SFT-PHYS-COSMO-REDSHIFT-001, SFT-PHYS-COSMO-DISTANCE-001, SFT-PHYS-COSMO-EXPANSION-001, SFT-PHYS-COSMO-STRUCTURE-GROWTH-001, SFT-PHYS-COSMO-BACKGROUND-001, SFT-PHYS-COSMO-DARK-BARYON-FRACTION-001, SFT-PHYS-GRAVITY-WAVE-001, SFT-PHYS-GRAVITY-HORIZON-001, SFT-PHYS-STELLAR-GALACTIC-TIDAL-TERMINAL-067, SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069, SFT-PHYS-COMPACT-HORIZON-THERMODYNAMICS-TERMINAL-071, SFT-CHEM-ELEM-ELEMENT-001, SFT-CHEM-STOICH-CONSERVATION-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ULTIMATE-FATE-001, SFT-ASTRO-ASTROBIOLOGY-001.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by the registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Closure is depth-independent for positive finite extension; empirical correspondence remains source, object, place, epoch, instrument and protocol bounded.

The eight binary axes generate exactly 256 forms. The unique preserving form is:

```
law-instrument-earth-life-social-boundary-carrier__branch-ownership-bounded__typed-br
anch-handoff-relation__source-owner-dependency-evidence-and-use-held__evidence-class-
explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-rows__no-
extra-rule
```

The survivor retains, in order, the complete carrier, explicit observational boundary, generated relation, reconstructible record, distinct evidence class, root-bound provenance, positive finite extension and no extra rule. Any candidate differing on at least one coordinate loses the corresponding required distinction and is eliminated. Thus 255 alternatives fail and one survives without a fitted choice.

Closure: `depth_independent`. Minimality: `True`. Named-form uniqueness: `True`.

Falsification and controls

Falsification condition: Reject if an instrument selects a law or another branch's object is silently reowned.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

External evidence

Evidence class: `source_bound_observation_archive_or_primary_measurement / primary_archive_or_measurement_correspondence`. Captured sources: 2; unresolved transports retained: 1. The post-seal consequence matched the registered structural label: `True`. External evidence selected the survivor: `false`.

- NASA-ADS-001 - transport failed_preserved; snapshot unresolved.
- NASA-LAMBDA-001 - transport captured; snapshot
sha256:7d4c758a1b6594a28da4b4478e0219a0e11fbb4dd0ba204449d25fc9204eda17.
- STSCI-MAST-001 - transport captured; snapshot
sha256:6d0a17e43d19504d059c0915fc90f8c842d878472878ce09b28a7e883c496701.

What it means

This law makes **astronomy scientific handoffs** admissible only when its carrier, relation, record and evidence boundary remain jointly reconstructible. It prevents a label, fitted output, favourable object or opaque model from replacing the scientific chain. The result may be extended with new finite sources and stronger evidence, but only by preserving every prior row and passing the same common engine.

Machine certificate

Derivation seal: sha256:c6b99ac95649612e4f1407601647e18bc33a4ca9bf09bbfbbeee525448813f69.

Independent certificate:

sha256:4df89f0be221973480f47e417d1561e932f93827b5a8070ceebfe15a09e3d5dc. Engine receipt:
sha256:1d8b4042aa2faf71d32266ffcd173e2d2462772695741ae0f290d55d0ba6a883. Candidate count:
256. Survivor count: 1.

Historical reconciliation

The V1/V2 census contains 763 entries. Every entry was reviewed. The 53 Astronomy-owned questions were registered as questions before the V3 derivation and mapped to admitted present claims. Prior answers were not used as premises. Universal Physics questions were preserved as typed handoffs rather than relabelled Astronomy claims.

Limits and extension-open status

This foundation does not claim that every astronomical catalogue row has been ingested, that every contingent initial condition is forced by structure, or that every instrument systematic has vanished. It claims something more precise: the foundational forms required to state, compare and falsify astronomical claims have been generated and admitted at the declared boundary, and all known V1/V2 Astronomy questions are accountable. New observations, objects, methods and lawful derivations are invited. A new result must extend the trace; it may not rewrite failed transports, adverse rows or dependency ownership.

Full-field continuation roadmap

- complete positional and time-domain catalogues.
- complete stellar and planetary populations.
- complete interstellar and chemical-evolution observations.
- complete galaxy and large-scale surveys.
- complete compact/high-energy/multimessenger catalogues.
- complete precision cosmic-history and early-universe evidence.
- complete astrobiological and observatory handoffs.

Each extension will be decomposed into explicit obligations, sealed before outcomes, admitted claim by claim and followed by a versioned paper update. Astronomy remains current-evidence closed at its foundation and open to discovery.

Reproduction

This Astronomy closure passed six focused tests, 72 sequential engine admissions, 72 independent validators and the branch integration audit. After all fifteen foundations were integrated, the untouched repository command `python3 -m sft verify-all` passed on 28 July 2026: 1,319/1,319 derivations replayed, 969/969 unit and end-to-end tests passed, 1,264/1,264 core executable lines covered, 1,011 empirical claims audited, 114/114 formal Physics claims reaching measurement and 5/5 live exact NIST/CODATA checks. The measured runtime was 4,753.740 seconds.

Rights, participation and scientific admission

Open criticism is welcome and does not require credentials. Scientific admission requires the shared protocol. Contributors may use the Ernos Labs designation only while preserving the theorem, constraints, complete enumeration, adverse evidence, immutable engine authority, retained authorship and open licenses. A fork that weakens those conditions remains free software but is not an Ernos Labs scientific admission.

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Discord: <https://discord.gg/ucwGryVxGr>

GitHub: <https://github.com/MettaMazza>

Paper license: CC BY 4.0

Code license: Apache-2.0

Machine identities

- Inventory: sha256:466048a43ba8e3fc91f33f231f18571bf80a3d16e934a18ec060678e814421cd
- Prior audit: sha256:4dbda9dcdf66a0d9e23605be4bfd41e2a5c28cd221989cac6d0d75192e322463
- Integration audit:
sha256:769af612334b3d79346b79430acb82b731bf8174f5cce60511773f88b991e9d4
- Source registry: sha256:b5d65c48dbbb538cb3ce95a9e57f3f7ad45d0e3c821013285a48b723a51a7126

- External target census:
sha256:76a74ee67434d0e774624ca6fdac747dc729a6578c1dfd0081c1b2636364d130
- Engine: sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a
- Verification authority:
sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8

Complete current-census successor supplement

The authoritative current scope of this paper is 77 live model-admitted claims (`astronomy_cosmology 77`). Counts retained inside the preceding-version narrative describe the dated freeze at which those passages were written; this successor table and the machine census control the current total.

The preceding manuscript did not contain 5 later-admitted claim section(s). They are included below in dependency order so the successor covers the complete current branch surface.

Solar/radio unit magnitude-frequency release

Claim ID: SFT-ASTRO-SOLAR-RADIO-UNIT-RELEASE-002

Exact statement: Every binary increase in release size has the exact inverse binary frequency share, so size times frequency completes One at every generated depth.

Dependency route: SFT-EARTH-TIPPING-FOLD-LOCK-002 ->

SFT-EARTH-QUAKE-MAGNITUDE-FREQUENCY-001 -> SFT-ASTRO-TRANSIENT-001 ->

SFT-PHYS-WAVE-PROPAGATION-001

Candidate generator: Generate the complete eight-axis product for SFT-ASTRO-SOLAR-RADIO-UNIT-RELEASE-002 and reconstruct its exact witness independently.

Grammar boundary: All positive finite binary depths with nonunit, fitted and incomplete-catalog controls.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: exact-positive-fold-parts-and-counts__binary-size-inverse-frequency-unit-product__normalized-law-measured-correspondence-separate__complete-source-and-adverse-record__complete-declared-product__derivation-seal-before-target__favorable-adverse-absent-heterogeneous-unresolved__no-extra-rule

Exact result: At every positive generated binary depth k , size ratio 2^k and frequency share $1/2^k$ multiply to One. The exact magnitude-frequency exponent is One; external solar and radio catalogs test its domain and departures.

Closure: depth_independent; minimality True; named-shape uniqueness True

External status: independently_replicated

Engine receipt: sha256:edf91022ff2d28e6865c002825eb0e508f2215b912e2bd66b5b161cbe0272958 at receipts/engine/model_admitted/SFT-ASTRO-SOLAR-RADIO-UNIT-RELEASE-002-edf91022ff2d28e6.json

Control	Passed	Expected	Observed
false_promise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Atomic Fold burst completion

Claim ID: SFT-ASTRO-ATOMIC-BURST-COMPLETION-002

Exact statement: The least nonempty release carrier half-One completes to One in exactly one Fold with no generated intermediate state.

Dependency route: SFT-ASTRO-SOLAR-RADIO-UNIT-RELEASE-002 -> SFT-ASTRO-PERIOD-TRANSIENT-001
-> SFT-FOUNDATION-FOLD-DYNAMICS-001

Candidate generator: Generate the complete eight-axis product for SFT-ASTRO-ATOMIC-BURST-COMPLETION-002 and reconstruct its exact witness independently.

Grammar boundary: The least positive binary release path with partial, multi-step and dimensional-overreach controls.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: exact-positive-fold-parts-and-counts__half-one-to-one-single-fold-release__normalized-law-measured-correspondence-separate__complete-source-and-adverse-record__complete-declared-product__derivation-seal-before-target__favorable-adverse-absent-heterogeneous-unresolved__no-extra-rule

Exact result: Half-One Folds to One in one exact step. No intermediate exists in that registered binary transition. This forces an atomic release class while leaving dimensional burst energy, duration and source structure to measurement.

Closure: depth_independent; minimality True; named-shape uniqueness True

External status: independently_replicated

Engine receipt: sha256:caedea69c14b9220fea53c58cefabbac44224e2dbceb634737c9cb072b521b66 at receipts/engine/model_admitted/SFT-ASTRO-ATOMIC-BURST-COMPLETION-002-caedea69c14b9220.json

Control	Passed	Expected	Observed
false_promise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Depth-seven planetary binary ladder

Claim ID: SFT-ASTRO-PLANETARY-BINARY-LADDER-002

Exact statement: The normalized depth-seven orbital carrier is generated by seven exact doublings from one-one-hundred-twenty-eighth to One.

Dependency route: SFT-ASTRO-ATOMIC-BURST-COMPLETION-002 -> SFT-ASTRO-PLANETARY-SYSTEM-001 -> SFT-ASTRO-ORBIT-001 -> SFT-MATH-EXACT-ARITHMETIC-001

Candidate generator: Generate the complete eight-axis product for SFT-ASTRO-PLANETARY-BINARY-LADDER-002 and reconstruct its exact witness independently.

Grammar boundary: The complete normalized depth-seven binary ladder and all adjacent-ratio, missing-body and nonuniversal-system controls.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: exact-positive-fold-parts-and-counts__depth-seven-exact-doubling-ladder__normalized-law-measured-correspondence-separate__complete-source-and-adverse-record__complete-declared-product__derivation-seal-before-target__favorable-adverse-absent-heterogeneous-unresolved__no-extra-rule

Exact result: The exact normalized ladder 1/128, 1/64, 1/32, 1/16, 1/8, 1/4, 1/2, One has adjacent ratio two throughout. It defines a binary-spacing architecture, not a universal fit to every planetary system.

Closure: depth_independent; minimality True; named-shape uniqueness True

External status: independently_replicated

Engine receipt: sha256:316542613f55d9f07e05141976cc889c47c787a61886f0aee26f6612db844a4e at receipts/engine/model_admitted/SFT-ASTRO-PLANETARY-BINARY-LADDER-002-316542613f55d9f0.json

Control	Passed	Expected	Observed
false_promise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

One-Fold lithium-seven depletion mechanism

Claim ID: SFT-ASTRO-LITHIUM-SEVEN-ONE-FOLD-DEPLETION-002

Exact statement: One exact stellar-processing Fold halves the prior three-sixteenths lithium-seven carrier to three-thirty-seconds and doubling restores the prior carrier.

Dependency route: SFT-ASTRO-PLANETARY-BINARY-LADDER-002 -> SFT-ASTRO-PRIMORDIAL-ABUNDANCE-001 -> SFT-PHYS-STELLAR-NUCLEAR-COLLAPSE-TERMINAL-069 -> SFT-CHEM-ELEM-ELEMENT-001

Candidate generator: Generate the complete eight-axis product for SFT-ASTRO-LITHIUM-SEVEN-ONE-FOLD-DEPLETION-002 and reconstruct its exact witness independently.

Grammar boundary: The exact three-sixteenths carrier under every positive finite binary processing depth, with mismatch and heterogeneous-history controls.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: exact-positive-fold-parts-and-counts__three-sixteenths-halved-to-three-thirty-seconds__normalized-law-measured-correspondence-separate__complete-source-and-adverse-record__complete-declared-product__derivation-seal-before-target__favorable-adverse-absent-heterogeneous-unresolved__no-extra-rule

Exact result: The prior lithium-seven carrier 3/16 multiplied by half-One is exactly 3/32, and one doubling reconstructs 3/16. The law is a zero-parameter one-stage depletion mechanism; observed abundance ratios and multi-stage stellar histories remain complete empirical tests.

Closure: depth_independent; minimality True; named-shape uniqueness True

External status: independently_replicated

Engine receipt: sha256:f295654ffdde16ad962d59c6af7ed9808f82b902a78f1c55785e01acea42ccb4 at receipts/engine/model_admitted/SFT-ASTRO-LITHIUM-SEVEN-ONE-FOLD-DEPLETION-002-f295654ffdde16ad.json

Control	Passed	Expected	Observed
false_promise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Complete post-seal Earth/Astronomy prior comparison

Claim ID: SFT-ASTRO-VALIDATION-PRIOR-COMPLETE-FAMILY-002

Exact statement: The sealed five-law family and five protected exact dependencies are compared with separately registered Earth, solar, planetary, galactic, abundance and transient observations while preserving every result class.

Dependency route: SFT-ASTRO-LITHIUM-SEVEN-ONE-FOLD-DEPLETION-002 ->
SFT-PHYS-PARKER-PROTON-ENERGY-TERMINAL-028 -> SFT-ASTRO-TULLY-FISHER-001 ->
SFT-PHYS-VALIDATION-GRAVITATIONAL-WAVE-CHIRP-RINGDOWN-074 ->
SFT-PHYS-MEAS-TARGET-CUSTODY-001

Candidate generator: Generate the complete eight-axis product for
SFT-ASTRO-VALIDATION-PRIOR-COMPLETE-FAMILY-002 and reconstruct its exact witness independently.

Grammar boundary: The sealed family against distinct identity-first sources with complete result-class retention.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: exact-positive-fold-parts-and-counts__sealed-earth-astronomy-family-versus-reg
istered-observations__normalized-law-measured-correspondence-separate__complete-sourc
e-and-adverse-record__complete-declared-product__derivation-seal-before-target__favor
able-adverse-absent-heterogeneous-unresolved__no-extra-rule

Exact result: The complete comparison tests normalized tipping structure, unit magnitude-frequency behavior, atomic transients, binary planetary spacing and one-stage lithium depletion alongside the protected Parker, Tully-Fisher, earthquake and ringdown results. Agreement, disagreement, heterogeneity, absence and unresolved rows all remain explicit.

Closure: depth_independent; minimality True; named-shape uniqueness True

External status: empirically_tested_and_independently_replicated

Engine receipt: sha256:22c0111a58263f844a79e781e80a54bc4f52412aa86f3519b2fa9ba83b344b27 at r
eceipts/engine/model_admitted/SFT-ASTRO-VALIDATION-PRIOR-COMPLETE-FAMILY-002-22c0111a
58263f84.json

Control	Passed	Expected	Observed
false_promise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Preceding-version abstract preserved for chronology

The following abstract is reproduced from version 1.0.0. Its counts and status language describe that dated version and do not override the current abstract or census above.

Astronomy is often presented as a catalogue of distant things and cosmology as a fit among inherited equations. This reconstruction begins lower. From the One and the Fold, and only through already admitted mathematical, informational, computational, physical, chemical and Earth dependencies, it asks what must be retained for any statement about a source beyond the observer to be distinguishable, reconstructible and falsifiable. The result is a foundational branch spanning source geometry, object identity, radiation, motion, time, stars, planetary systems, interstellar matter, galaxies, compact objects, multimessenger records, expansion, early-universe evidence, dark-sector inference and scientific handoffs. Each law is zero-parameter in its proof layer: no fitted constant, target value, irrational or imaginary proof number, conventional continuum or semantic numerical zero selects a survivor. Dimensional observations remain exact source records outside that proof layer. This paper documents the entire forward chain, the candidate grammars, the controls, the empirical boundary, the adverse result, the independent certificates and the open continuation programme.

Registered source surface

Human-readable scholarly references remain in the inherited paper. Machine source IDs, custodians, releases, locators, source captures, access outcomes, hashes and transport status remain in the current claim packages and external-source registries. Source prestige is not evidential authority; each source is used only in its registered role. The Lean source-binding gate checks custody and identity, not empirical truth by reputation.

Limitations and open frontier

The current result is formally complete only to the named dated grammar, census and artifact graph. Native Lean theoremhood is presently established for the operational root and generic gate implications; the other scientific claims are artifact checked rather than individually restated as Lean propositions. Empirical validation remains claim specific, and adverse, unavailable and unresolved evidence remains open where the current records say so. New discoveries, falsifications, stronger comparisons and native claim-level formalisations may enter only through a later version without rewriting this one.

Data and code availability

The successor Markdown is

`publications/successors/astronomy_cosmology/FROM_ONE_SKY_TO_COSMOS_PAPER_001_V1_1.md`
and its scientific predecessor is

`publications/current/astronomy_cosmology/FROM_ONE_SKY_TO_COSMOS.md`. The Lean sources and report are under `generated/lean4_validation/`; the report is

`generated/lean4_validation/reports/whole_model_validation.json`. Current claims are indexed by `census/claims.json`; complete candidates, decisions, controls, certificates, observations and receipts remain under `claims/<claim-id>/` and `receipts/`. The paper's evidence map and draft deposit metadata sit beside the successor source. No remote action was performed.

References and source registry

1. Smith, Maria. The preceding version of this paper, preserved at the repository path identified above, 2026.
2. Smith, Maria. *There Is No Nothing*. Ernos Labs Methods Paper 00, versioned publication series, 2026.
3. de Moura, Leonardo, and Sebastian Ullrich. "The Lean 4 Theorem Prover and Programming Language." *Automated Deduction - CADE 28*, 2021. DOI: [10.1007/978-3-030-79876-5_37](https://doi.org/10.1007/978-3-030-79876-5_37).
4. Lean FRO. *The Lean Language Reference*. lean-lang.org/doc/reference/latest.
5. Ernos Labs. Current claim packages, source registries and Lean verification report, repository snapshot bound by the machine identities below.

Successor machine-identity appendix

- Preceding source: `publications/current/astronomy_cosmology/FROM_ONE_SKY_TO_COSMOS.md`
- Preceding source SHA-256:
`sha256:135b2ada93caf788571b9e7c0d384d60ec0851d9c47a85fe432d6ad2618ef643`
- Lean report: `generated/lean4_validation/reports/whole_model_validation.json`
- Lean report SHA-256:
`sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf`
- Ordered census SHA-256:
`sha256:f3f540f77a9b677bf7ddb9f5c6ea1ea41f08e57bb07163e25f385fd4d1dcde71`
- Execution manifest SHA-256:
`sha256:517fd288f4484f43fdc0343b17fcabd06ea9e12c977a4512d4a891dd038fce41`
- Protected engine seal:
`sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a`

- Verification-authority seal:
sha256:bfb810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8
- Publication authorised: false
- Remote actions performed: false

Lean 4 independent verification update

Verified result

The pinned Lean toolchain `leanprover/lean4:v4.32.0` compiled the verification project and the executable verifier returned `PASS`. Its immutable report identity for this successor is

sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf.

Verified surface	Result
Ordered census claims	2,777
Proof-bearing accepted claim gates	2,777
Source-bound claims	2,777
Candidate records	898,902
Decision records	898,902
Passed controls	11,108
Registered branches	17
Issues	0

This paper's registered branch surface contributes **77** of the accepted claims.

Native theorem proof and artifact verification are different layers

`SFTValidation/Root.lean` formalises the exact registered two-member operational grammar: `unpresentedAbsence` and `presentedOccurrence`. The decision function rejects the former and accepts the latter. Lean proves existence by the accepted constructor and uniqueness by exhaustive case analysis over the two constructors. `#print axioms` reports no imported or user-declared axioms for the exported root theorems. This is a kernel-checked proof of the unique survivor inside the exact registered operational grammar.

`SFTValidation/Gates.lean` defines twelve Boolean acceptance obligations and can construct an inhabitant of `ClaimGate.Accepted gate` only when all twelve equal `true`. The runtime verifier then parses the complete ordered census and execution manifest, recomputes identities, cardinalities, one-to-one decision coverage, survivor count, controls, closure fields, empirical boundaries, certificate bindings and authoritative receipts, and calls that proof-bearing constructor for every current claim.

The exact scientific statements of the other claims are not all re-expressed as 2,776 separate Lean propositions in this version. They are validated as the repository's complete registered machine artifacts through Lean-executed checks. The source-manifest gate uses a read-only Python bridge solely to instantiate the already registered source factories; Lean consumes the result. This layer does not replace the protected admission engine, does not write receipts and does not substitute for source experiments or empirical replication.

Fail-closed custody event

The first whole-model run halted on six source-capture byte hashes. The discrepancy was traced to line-ending normalisation in the working tree, not to a changed scientific target or a failed theorem. The registered source bytes were restored, exact-byte Git attributes were added for those six captures, all 385 registered external source bindings were re-audited with no mismatch, and the unchanged Lean verification was rerun to `PASS`. The preserved halt demonstrates that source drift is detected rather than silently forgiven.

What the result supports

The result materially strengthens the model's case for formal coherence, exhaustive current-census coverage, unique-survivor enforcement, dependency and provenance integrity, cross-branch consistency and reproducibility by an implementation outside the admission engine. It does not by itself establish that every empirical claim is true in nature, that the registered grammar is the only conceivable grammar outside its stated boundary, or that SFT is superior to every rival theory. Those questions remain governed by the papers' declared experiments, falsification conditions and comparative tests.

Successor conclusion

Version 1.1.0 preserves the preceding paper's derivations, evidence classes, corrections, adverse and unresolved records, ownership limits and open frontier, while integrating the independent Lean 4 result and every later current claim in its declared scope. This paper contributes 77 current claim(s) within `astronomy_cosmology` to the total dependency spine.

The new formal result establishes that the registered two-class operational root has one Lean-proved survivor and that all 2,777 current claims satisfy the proof-bearing artifact gates. The major conflation resolved by this update is between native theorem proof and whole-repository artifact verification: both are valuable, but they are not the same evidential act. No empirical status is strengthened merely because the artifact graph passes, and no historical halt or correction is erased.

The current evidence therefore establishes formal coherence, current-census completeness, unique-survivor enforcement and intact source, dependency, certificate and receipt custody for this version. Field-specific empirical conclusions remain exactly those authorised by their current records. Lawful discovery, falsification, stronger evidence and versioned extension remain open, and publication itself remains pending Maria Smith's explicit confirmation.